Hausdorff spaces: finite case, metrizability, and uniqueness of limits

Problem. Recall that a topological space (X, τ) is *Hausdorff* (or T_2) if for all distinct $x, y \in X$ there exist open sets $U, V \in \tau$ with $x \in U$, $y \in V$, and $U \cap V = \varnothing$.

(a) Show that if X is finite there is only one topology (up to homeomorphism) on X which is Hausdorff.

(b) Show that if (X, τ) is metrizable then it is Hausdorff.

(c) Show that if (X, τ) is Hausdorff and (x_n) is a convergent sequence in X, then its limit is unique.

Facts.

- In a Hausdorff space, every singleton is closed; i.e. $T_2 \Rightarrow T_1$.
- If X is finite and T_1, then every subset of X is closed (finite union of singletons), hence open; thus τ is the discrete topology.
- If d is a metric on X and $x \neq y$, then with $r = \frac{1}{2}d(x, y)$ the balls $B_r(x)$ and $B_r(y)$ are disjoint; hence every metric space is Hausdorff.

Solution to (a). Let X be finite and Hausdorff. Then X is T_1, so each singleton $\{x\}$ is closed.

Since X is finite, any subset $A \subseteq X$ is a finite union of singletons and hence closed.

Therefore every subset is closed and its complement is open; equivalently, every subset is open.

Thus τ is the discrete topology. Any two finite discrete spaces of the same cardinality are homeomorphic (every bijection is a homeomorphism), so there is only one Hausdorff topology on X *up to homeomorphism*.

Solution to (b). Let (X, d) be a metric space and take $x \neq y \in X$. Set $r = \frac{1}{2}d(x, y) > 0$. Then $B_r(x) \cap B_r(y) = \varnothing$. Hence X is Hausdorff.

Solution to (c). Suppose $(x_n) \to x$ and $(x_n) \to y$ in a Hausdorff space with $x \neq y$. Choose disjoint open neighborhoods $U \ni x$ and $V \ni y$. By convergence, there exists N_1 with $x_n \in U$ for all $n \geq N_1$, and N_2 with $x_n \in V$ for all $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$ we would have $x_n \in U \cap V = \varnothing$, a contradiction. Hence the limit is unique.

Examples.

- Hausdorff: $\mathbb{R}^n$ with the Euclidean metric; any discrete space; subspaces and products of Hausdorff spaces.
- T_1 but not Hausdorff: an infinite set with the cofinite topology (nonempty open sets always intersect).
- Non-Hausdorff with non-unique limits: the indiscrete topology $\{\varnothing, X\}$, where every sequence converges to every point.

Three Basic Invariance and Comparison Facts

Problem 2: Continuity of the Identity Between Two Topologies on the Same Set

Statement

Let $\mathrm{id} : (X, \tau) \to (X, \rho)$ be the identity on a fixed set X endowed with (possibly different) topologies τ and ρ. Is id necessarily continuous?

Theory

A map $f : (X, \tau) \to (Y, \sigma)$ is continuous iff $f^{-1}(U) \in \tau$ for all $U \in \sigma$. For the identity $\mathrm{id} : X \to X$ we have $\mathrm{id}^{-1}(U) = U$ for every $U \subseteq X$.

Answer and Proof

The map $\mathrm{id} : (X, \tau) \to (X, \rho)$ is continuous iff $\rho \subseteq \tau$. Indeed,

$$\text{id continuous} \iff \left(\forall U \in \rho,\ \mathrm{id}^{-1}(U) = U \in \tau\right) \iff \rho \subseteq \tau.$$

Hence *not necessarily*: id is continuous exactly when the codomain topology is *coarser* (or equal) to the domain topology.

Examples and Counterexamples

- If τ is discrete and ρ is indiscrete on X, then $\rho \subseteq \tau$; thus $\mathrm{id} : (X, \tau) \to (X, \rho)$ is continuous.
- If τ is indiscrete and ρ is discrete, then $\rho \nsubseteq \tau$; thus $\mathrm{id} : (X, \tau) \to (X, \rho)$ is *not* continuous.

- Symmetrically, $\mathrm{id} : (X, \rho) \to (X, \tau)$ is continuous iff $\tau \subseteq \rho$.

Problem 3: Hausdorffness and Second Countability are Topological Invariants

Statement

Show that the Hausdorff property and second countability are preserved under homeomorphisms.

Theory

A homeomorphism $f : X \to Y$ is a bijection such that both f and f^{-1} are continuous. Such a map carries open sets to open sets and bases to bases via images.

Hausdorff Invariance: Proof

Assume $f : X \to Y$ is a homeomorphism and X is Hausdorff. For $y_1 \neq y_2$ in Y, set $x_i = f^{-1}(y_i)$. Since X is Hausdorff there exist disjoint open sets $U_i \ni x_i$. Then $f(U_i)$ are disjoint open neighbourhoods of y_i, so Y is Hausdorff. The converse follows by applying the same argument to f^{-1}.

Second Countability Invariance: Proof

Let $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$ be a countable base for X. Because f is open and bijective, $\{f(B_n) : n \in \mathbb{N}\}$ is a countable base for Y. Conversely, apply the same reasoning to f^{-1}.

Example / Non-Example

$(\mathbb{R}, \text{usual})$ is Hausdorff and second countable, while $(\mathbb{R}, \text{cofinite})$ is T_1 but neither Hausdorff nor second countable. Therefore they are not homeomorphic, illustrating invariance.

Problem 4: Equality of the Euclidean Metric Topology and the Product Topology on $\mathbb{R}^n$

Statement

Show that the topology on $\mathbb{R}^n$ induced by the Euclidean metric equals the product topology inherited from $\mathbb{R}$ with its usual metric.

Theory

Let $\|\cdot\|_2$ be the Euclidean norm and $\|\cdot\|_\infty$ the sup-norm on $\mathbb{R}^n$. For all $v \in \mathbb{R}^n$,

$$\|v\|_\infty \leq \|v\|_2 \leq \sqrt{n}\,\|v\|_\infty.$$

Hence for balls centered at $x \in \mathbb{R}^n$ and $r > 0$,

$$B_2(x,r) \subseteq B_\infty(x,r) \subseteq B_2\Big(x, \sqrt{n}\,r\Big).$$

Basic product-open sets are the open boxes $\prod_{i=1}^n (a_i, b_i) = B_\infty(x,r)$.

Proof

Fix $x \in \mathbb{R}^n$ and $r > 0$.

- By $\|v\|_2 \leq \sqrt{n}\|v\|_\infty$, we have $B_\infty(x, r/\sqrt{n}) \subseteq B_2(x,r)$. Since $B_\infty(x, r/\sqrt{n})$ is a basic product-open box, every Euclidean-open set is product-open.
- By $\|v\|_\infty \leq \|v\|_2$, we have $B_2(x,r) \subseteq B_\infty(x,r)$. Hence every product-open set is Euclidean-open.

Therefore the Euclidean metric topology and the product topology on $\mathbb{R}^n$ coincide. $\square$

Example

In $\mathbb{R}^2$, each open rectangle $(a,b) \times (c,d)$ is Euclidean-open, and every Euclidean open disk contains and is contained in some open rectangle; thus both topologies agree.

Product Topology: Definition, Properties, Examples, and Counterexamples

Definition (Product Topology)

Let $\{(X_i, \tau_i)\}_{i\in I}$ be topological spaces and let $X = \prod_{i\in I} X_i$ with projections $\pi_i : X \to X_i$.

Initial-topology definition. The *product topology* τ on X is the coarsest topology for which every π_i is continuous. Equivalently, τ is generated by the subbasis

$$\mathcal{S} = \{\ \pi_i^{-1}(U)\ :\ i \in I,\ U \in \tau_i\ \}.$$

Basic-open boxes (finite support). A basis for the product topology on $X = \prod_{i\in I} X_i$ is

$$\mathcal{B}\ =\ \left\{\ \prod_{i\in I} U_i\ :\ U_i \in \tau_i \text{ and } U_i = X_i \text{ for all but finitely many } i \in I\right\}.$$

Equivalently, each element of $\mathcal{B}$ can be written as a finite intersection of subbasic cylinders,

$$\prod_{i\in I} U_i\ =\ \bigcap_{k=1}^{m} \pi_{i_k}^{-1}(U_{i_k}), \quad \text{where } \{i_1, \ldots, i_m\} \subset I \text{ and } U_{i_k} \in \tau_{i_k}.$$

Finite products. If $I = \{1, \ldots, n\}$ is finite, the condition "$U_i = X_i$ for all but finitely many i" is automatic. Thus the basic open sets are precisely the ordinary open rectangles

$$U_1 \times U_2 \times \cdots \times U_n, \qquad U_j \in \tau_j\ (1 \le j \le n).$$

Universal Property

For any space (Y, σ) and a map $f : Y \to \prod_{i\in I} X_i$,

$$f \text{ is continuous} \iff \pi_i \circ f : Y \to X_i \text{ is continuous for all } i \in I.$$

Basic Properties

- Each projection π_i is continuous and open.
- A net (x^α) in X converges to x iff $\pi_i(x^\alpha) \to \pi_i(x)$ in X_i for every $i \in I$.
- If every X_i is Hausdorff, then $\prod_{i\in I} X_i$ is Hausdorff.
- (**Tychonoff**) If every X_i is compact, then $\prod_{i\in I} X_i$ is compact.
- If I is countable and each X_i is second countable, then $\prod_{i\in I} X_i$ is second countable.

Examples

1. $\mathbb{R}^n \cong \prod_{i=1}^n \mathbb{R}$; the product topology equals the Euclidean topology.
2. The n-torus $\mathbb{T}^n = (S^1)^n$ with the product topology.
3. *Cantor space* $\{0,1\}^{\mathbb{N}}$ with product of the discrete topology on $\{0,1\}$: compact, totally disconnected, perfect; homeomorphic to the middle-thirds Cantor set.
4. *Baire space* $\mathbb{N}^{\mathbb{N}}$ with product of the discrete topology on $\mathbb{N}$ (Polish, non-compact).
5. Function spaces X^I as products: evaluation maps are the projections.

Counterexamples and Cautions

Product vs. Box topology (infinite products). The *box topology* on $\prod_{i\in I} X_i$ uses as a basis all boxes $\prod_{i\in I} U_i$ with $U_i \in \tau_i$ (no finiteness restriction). If I is infinite, the box topology is strictly finer than the product topology in general.

Sequence contrast in $\mathbb{R}^{\mathbb{N}}$. Let $x^{(m)} = (x_n^{(m)})_{n\in\mathbb{N}}$ with $x_m^{(m)} = 1$ and $x_n^{(m)} = 0$ for $n \neq m$. Then $x^{(m)} \to 0$ in the product topology (coordinatewise convergence), but $x^{(m)}$ does not converge to 0 in the box topology since the basic box $\prod_n(-\frac{1}{2}, \frac{1}{2})$ contains no $x^{(m)}$.

Cofinite not preserved. If each X_i is infinite with the cofinite topology, the product is not cofinite (indeed it is compact by Tychonoff but has many more open sets).

Second countability can fail for uncountable products. Even when all factors are second countable, $\mathbb{R}^I$ with I uncountable is not second countable.

Cofinite (Finite-Complement) Topology: Definition, Properties, Examples

Definition

Let X be a set. The *cofinite topology* on X is

$$\tau_{\text{cof}} = \{\varnothing\} \ \cup \ \{\, U \subseteq X : X \setminus U \text{ is finite}\,\}.$$

Equivalently, the closed sets are precisely the finite subsets of X together with X itself.

Basic Properties

1. T_1. Singletons are finite, hence closed. Therefore (X, τ_{cof}) is T_1.

2. **(Non-)Hausdorff.** If X is infinite, any two nonempty open sets have finite complements, so their intersection is nonempty. Thus the space is not Hausdorff. If X is finite, then every subset is open (the topology is discrete), so the space is Hausdorff.

3. **Compactness.** Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X. If some U_{α_0} is nonempty, then $F = X \setminus U_{\alpha_0}$ is finite; finitely many other opens cover F, giving a finite subcover.

4. **Connectedness (hyperconnected).** No two nonempty open sets are disjoint, hence there are no nontrivial clopen subsets.

5. **Countability properties.**
   - If X is countably infinite, then (X, τ_{cof}) is first countable and second countable. For $x \in X$, enumerate $X \setminus \{x\} = \{y_1, y_2, \dots\}$ and let $U_n = X \setminus \{y_1, \dots, y_n\}$; the family $\{U_n : n \in \mathbb{N}\}$ is a neighborhood base at x.
   - If X is uncountable, then (X, τ_{cof}) is not first countable (hence not second countable). Indeed, given any countable family $\{U_n\}$ of neighborhoods of x, the union of the finite complements $F_n = X \setminus U_n$ is countable; choose $y \in X \setminus (\{x\} \cup \bigcup_n F_n)$; then $X \setminus \{y\}$ is a neighborhood of x not containing any U_n.

6. **Closures and density.** Since the only proper closed sets are finite,

$$\overline{A} = \begin{cases} A, & A \text{ finite}, \\ X, & A \text{ infinite}. \end{cases}$$

   Hence every infinite subset is dense; in particular, the space is separable whenever X is infinite.

7. **Sequential convergence.** A sequence (x_n) converges to x iff for every $y \neq x$ the set $\{n : \ x_n = y\}$ is finite. Consequently, a sequence with pairwise distinct terms converges to every point of X.

Examples

- $(\mathbb{N}, \tau_{\mathrm{cof}})$: compact, T_1, first/second countable, not Hausdorff.
- $(\mathbb{R}, \tau_{\mathrm{cof}})$: compact, T_1, connected, neither first nor second countable, not Hausdorff.
- Any finite set with the cofinite topology is discrete (hence Hausdorff and compact).

Counterexamples / Cautions

- On an infinite X, (X, τ_{cof}) is not Hausdorff: no two nonempty opens are disjoint.
- If X is uncountable, the space is not first countable and thus not metrizable.
- Products of cofinite spaces (with the product topology) are compact by Tychonoff but are generally not cofinite and may fail first/second countability for large index sets.

Problem 5: Diagonals and Equality Sets

Let (X, τ) and (Y, ρ) be topological spaces with (Y, ρ) Hausdorff.

(a) The diagonal is closed in $Y \times Y$

Statement. Show that $\Delta = \{(y, y) : y \in Y\} \subset Y \times Y$ is closed (product topology).

Proof. Let $(y_1, y_2) \in (Y \times Y) \setminus \Delta$, so $y_1 \neq y_2$. Since Y is Hausdorff, there exist disjoint open sets $U \ni y_1$ and $V \ni y_2$ with $U \cap V = \varnothing$. Then

$$(y_1, y_2) \in U \times V \subset (Y \times Y) \setminus \Delta,$$

because for any $(u, v) \in U \times V$ we cannot have $u = v$ (that would force $u \in U \cap V = \varnothing$). Hence $(Y \times Y) \setminus \Delta$ is open and Δ is closed. □

Remark. The converse holds: Y is Hausdorff if and only if Δ is closed in $Y \times Y$.

(b) Equality set of two continuous maps is closed

Statement. Let $f, g : X \to Y$ be continuous. Show that $E = \{x \in X : \ f(x) = g(x)\}$ is closed in X.

Proof. Consider $F = (f, g) : X \to Y \times Y$, which is continuous in the product topology. Then

$$E = F^{-1}(\Delta).$$

By part (a), Δ is closed in $Y \times Y$, hence E is the preimage of a closed set under a continuous map and is therefore closed in X. □

Examples

- If $Y = \mathbb{R}$ and $f, g : X \to \mathbb{R}$ are continuous, then

  $$E = \{x : \ f(x) = g(x)\} = (f - g)^{-1}(\{0\}),$$

  which is closed since $\{0\}$ is closed and $f - g$ is continuous.
- On $X = \mathbb{R}$, $f(x) = x^3 - 1$ and $g(x) = x - 1$ give $E = \{1\}$, a closed set.

Counterexample without Hausdorffness

Let $Y = \{a, b\}$ with the indiscrete topology. Then Y is not Hausdorff and $\Delta \subset Y \times Y$ is not closed. Take any space X and a non-closed set $A \subset X$ (e.g. $X = \mathbb{R}$, $A = (0, 1)$). Define

$$f(x) \equiv a, \qquad g(x) = \begin{cases} a, & x \in A, \\ b, & x \notin A. \end{cases}$$

Every map into an indiscrete space is continuous, so f, g are continuous. But $E = \{x : f(x) = g(x)\} = A$, which is not closed. Thus the Hausdorff hypothesis in (b) is essential.

Diagonal and G_δ-Diagonal in Topology

Definitions

Let X be a topological space and $\Delta_X = \{(x, x) : x \in X\} \subseteq X \times X$ its diagonal.

- X is *Hausdorff* $\iff$ Δ_X is *closed* in $X \times X$.
- X has a *G_δ-diagonal* if there exist open sets $U_n \subseteq X \times X$ such that

$$\Delta_X = \bigcap_{n=1}^{\infty} U_n.$$

Basic Facts

1. **Metric spaces have G_δ-diagonals.** If (X, d) is metric, the function $(x, y) \mapsto d(x, y)$ is continuous on $X \times X$, hence

$$\Delta_X = \bigcap_{m=1}^{\infty} \Big\{(x, y) \in X \times X : d(x, y) < 1/m\Big\}$$

   is a G_δ subset of $X \times X$.

2. **First countable Hausdorff $\Rightarrow$ G_δ-diagonal.** If X is Hausdorff and for each $x \in X$ there is a countable neighborhood base $\{B_n(x)\}_{n \in \mathbb{N}}$, then

$$U_n = \bigcup_{x \in X} B_n(x) \times B_n(x) \qquad (n \in \mathbb{N})$$

   are open in $X \times X$ and $\Delta_X = \bigcap_n U_n$.

3. **Sneider's Theorem (compact case).** A compact Hausdorff space K is metrizable if and only if Δ_K is a G_δ in $K \times K$.

Examples

- $\mathbb{R}^n$ with its Euclidean topology: Hausdorff and metric $\Rightarrow \Delta$ is closed and a G_δ.
- Any compact metrizable space (e.g. the Cantor set, the n-torus $(S^1)^n$) has a G_δ-diagonal.
- The Niemytzki (Moore) plane: Hausdorff and first countable, hence has a G_δ-diagonal, but is not normal; consequently not metrizable.

Counterexamples / Cautions

- For uncountable I, the Tikhonoff cube $[0,1]^I$ is compact Hausdorff but not metrizable; hence by Sneider, $\Delta_{[0,1]^I}$ is *not* a G_δ in $[0,1]^I \times [0,1]^I$.
- The indiscrete space on $\{a, b\}$ is not Hausdorff; Δ is not closed (and not G_δ).
- The cofinite topology on an infinite set is T_1 but not Hausdorff; Δ is not closed (hence not G_δ).

Problem 6: Closure vs. Sequential Closure

Setup and Definitions

For $A \subseteq X$, the *sequential closure* $\mathrm{scl}(A)$ is the set of all limits of sequences (x_n) with $x_n \in A$. We say A is *sequentially closed* if $\mathrm{scl}(A) \subseteq A$, i.e. whenever $x_n \in A$ and $x_n \to x$ in X, then $x \in A$.

(a) Co-countable topology: sequentially closed $\not\Rightarrow$ closed

Assume X is uncountable and $\tau = \{\varnothing\} \cup \{\, U \subseteq X : \ X \setminus U \text{ is countable} \,\}$ (the co-countable topology).

Lemma 0.1. *In (X, τ), a sequence (x_n) converges to x if and only if it is eventually constant equal to x.*

Proof. ($\Rightarrow$) Let $(x_n) \to x$. Any neighborhood of x is of the form $X \setminus C$ with C countable and $x \notin C$; thus for each such C, $\{n : \ x_n \in C\}$ is finite. Let S be the set of values taken by the tail of (x_n); S is countable. If $x \notin S$, take $C = S$ and get a contradiction. Hence $x \in S$. Taking $C = S \setminus \{x\}$ forces $x_n \in \{x\}$ eventually. ($\Leftarrow$) If $x_n = x$ eventually, then every neighborhood of x contains all sufficiently large n, so $x_n \to x$. $\square$

Proposition 0.2. *Every subset $A \subseteq X$ is sequentially closed, but an uncountable proper subset A is not closed.*

Proof. If $x_n \in A$ and $x_n \to x$, the lemma shows $x_n = x$ eventually, hence $x \in A$; thus A is sequentially closed. Closed sets in the co-countable topology are precisely the countable sets together with X; therefore any uncountable proper A is not closed. $\square$

Example. On $X = \mathbb{R}$ with the co-countable topology, $A = \mathbb{R} \setminus \mathbb{Q}$ is sequentially closed but not closed.

(b) Second countable spaces: sequentially closed $\Leftrightarrow$ closed

Proposition 0.3. *If (X, τ) is second countable, then for every $A \subseteq X$ we have* $\operatorname{scl}(A) = \overline{A}$*. In particular, A is sequentially closed iff A is closed.*

Proof. Let $\mathcal{B} = \{B_1, B_2, \dots\}$ be a countable base for X. Fix $x \in X$ and enumerate $\mathcal{B}_x = \{B \in \mathcal{B} : x \in B\}$ as $(C_n)_{n \geq 1}$. Define $U_n = \bigcap_{k=1}^{n} C_k$. Then (U_n) is a decreasing countable neighborhood base at x. If $x \in \overline{A}$, then $A \cap U_n \neq \varnothing$ for all n; choose $x_n \in A \cap U_n$. By construction $x_n \to x$. Hence every point of $\overline{A}$ is a sequential limit of points of A, so $\operatorname{scl}(A) \supseteq \overline{A}$. The reverse inclusion is always true. Therefore $\operatorname{scl}(A) = \overline{A}$, and the equivalence follows. $\square$

Examples. In $(\mathbb{R}, \text{usual})$ the set $A = (0, 1)$ is not sequentially closed since $x_n = 1/n \to 0 \notin A$. In contrast, in the co-countable $\mathbb{R}$ every subset is sequentially closed, while uncountable proper subsets are not closed.

Problem 6: Extra Examples and Sequential Spaces

More Examples for (a): Co-countable Topology (Uncountable X)

In the co-countable topology on an uncountable set X, a sequence converges iff it is eventually constant. Hence every subset $A \subseteq X$ is sequentially closed, while the closed sets are exactly the countable subsets and X itself.

- $X = \mathbb{R}$, $A = \mathbb{R} \setminus \mathbb{Q}$: sequentially closed, not closed.
- $X = \mathbb{R}$, $A = (0, 1) \cup \{\pi\}$: sequentially closed, not closed.
- X uncountable, $A = X \setminus C$ with nonempty countable C: sequentially closed, not closed.
- Any countably infinite $C \subset X$ is closed; its sequential closure equals C.

Non-example. If X is finite with the co-countable topology, then X is discrete; sequentially closed sets are exactly the closed sets.

More Examples for (b): Second Countable Spaces

In every second countable space (hence first countable), A is sequentially closed iff A is closed.

- $\mathbb{R}^n$: $(0,1)$ is not sequentially closed (since $1/n \to 0 \notin (0,1)$); $[0,1]$ is sequentially closed (closed).
- Any metric space (Polish spaces, manifolds, graphs with path metric, metric CW-complexes): same equivalence.
- Subspaces of second countable spaces: e.g. $S^1 \subset \mathbb{R}^2$. The set $S^1 \setminus \{1\}$ is not sequentially closed (points approach 1).
- Countable products of second countable spaces (e.g. $\mathbb{R}^{\mathbb{N}}$ with product topology): still second countable; sequentially closed $\Leftrightarrow$ closed.

Non-examples. For uncountable I, $\mathbb{R}^I$ (product topology) is not second countable; there are subsets whose topological closure strictly contains their sequential closure. Likewise, $\mathbb{R}$ with the co-countable topology is not first/second countable; every set is sequentially closed but many are not closed.

Sequential Spaces (Sequential Topology)

Definition 0.4. A topological space X is *sequential* if for all $A \subseteq X$,

$$\overline{A} = \operatorname{scl}(A),$$

i.e. A is closed iff A is sequentially closed.

Facts and Examples.

- Every first countable space (in particular, every metric or second countable space) is sequential.
- The Arens–Fort space is sequential but not first countable.
- Quotients of metric spaces are sequential.
- Products of sequential spaces need not be sequential.

Non-sequential Counterexamples.

- The Tikhonoff cube $[0,1]^I$ with I uncountable is not sequential.
- The Stone–Čech compactification $\beta\mathbb{N}$ is not sequential.
- The Fortissimo space is not sequential (fails countable tightness).
- $\mathbb{R}$ with the co-countable topology is not sequential (every set is sequentially closed, many are not closed).

Sequentially Continuous vs. Continuous

Definition 0.5. Let $f : X \to Y$ be a map between topological spaces.

- f is *sequentially continuous at* $x \in X$ if for every sequence (x_n) with $x_n \to x$ in X we have $f(x_n) \to f(x)$ in Y.
- f is *continuous* if $f^{-1}(U)$ is open in X for every open $U \subseteq Y$.

Proposition 0.6. *If X is a sequential space (e.g. first countable, metric, or second countable), then $f : X \to Y$ is continuous iff it is sequentially continuous.*

Example 0.7 (Sequentially continuous but not continuous)**.** Let $X = \mathbb{R}$ with the co-countable topology and let $Y = \{0,1\}$ with the discrete topology. Define

$$f(x) = \begin{cases} 0, & x \in \mathbb{Q}, \\ 1, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

In X, a sequence converges to x iff it is eventually constant equal to x. Hence for any $x_n \to x$, the sequence $f(x_n)$ is eventually constant equal to $f(x)$, so f is sequentially continuous. However, $\{0\} \subset Y$ is open while $f^{-1}(\{0\}) = \mathbb{Q}$ is not open in the co-countable topology. Thus f is not continuous.

Problem 8: Gluing (Pasting) Continuous Maps

Let (X,τ) be a topological space with $X = A \cup B$ and (Y,ρ) another space. Let $g : A \to Y$ and $h : B \to Y$ be continuous (with subspace topologies) and suppose $g = h$ on $A \cap B$. Define

$$f(x) = \begin{cases} g(x), & x \in A, \\ h(x), & x \in B. \end{cases}$$

(a) If A and B are closed, then f is continuous

Proof. Let $C \subseteq Y$ be closed. Since g is continuous, $g^{-1}(C)$ is closed in A, so $g^{-1}(C) = A \cap F$ for some closed $F \subseteq X$. Similarly $h^{-1}(C) = B \cap G$ for some closed $G \subseteq X$. Then

$$f^{-1}(C) = \Big(A \cap g^{-1}(C)\Big) \cup \Big(B \cap h^{-1}(C)\Big) = (A \cap F) \cup (B \cap G),$$

a union of closed subsets of X. Hence $f^{-1}(C)$ is closed for every closed C, so f is continuous. □

(b) If A is not closed, continuity may fail

Counterexample. Let $X = (0, 2]$ with the subspace topology from $\mathbb{R}$. Take $A = (0, 1)$ (not closed in X) and $B = [1, 2]$ (closed).

Define $g \equiv 0$ on A and $h \equiv 1$ on B. They agree on $A \cap B = \varnothing$, so the glued map

$$f(x) = \begin{cases} 0, & x \in A, \\ 1, & x \in B \end{cases}$$

is well-defined.

For the open set $U = \left(\frac{1}{2}, \frac{3}{2}\right) \subset \mathbb{R}$ we have

$$f^{-1}(U) = B = [1, 2],$$

which is not open in X (every neighborhood of 1 in X meets A).

Thus f is not continuous.

Remark (Open-cover version). If A and B are both *open* in X, then for any open $U \subseteq Y$,

$$f^{-1}(U) = (A \cap g^{-1}(U)) \cup (B \cap h^{-1}(U)),$$

and each term is open in X (since $g^{-1}(U)$ is open in A, it equals $A \cap G$ for some open $G \subseteq X$). Hence f is continuous when A, B form an open cover as well.

Problem 9: Finite Products of Metrizable Spaces

Statement. Show that a finite product of metrizable topological spaces is metrizable. (Hint: start with two metric spaces and define a metric on the product.)

Proof for two factors. Let (X, d) and (Y, e) be metric spaces. Set $d_1(x, x') = \min\{1, d(x, x')\}$ and $e_1(y, y') = \min\{1, e(y, y')\}$, which induce the same topologies as d and e but are bounded by 1. Define

$$D\Big((x, y), (x', y')\Big) = d_1(x, x') + e_1(y, y').$$

Then D is a metric on $X \times Y$.

We claim that the topology induced by D equals the product topology.

(i) If $D((x, y), (x', y')) < r$, then $d_1(x, x') < r$ and $e_1(y, y') < r$, hence

$$B_D((x, y), r) \subseteq B_{d_1}(x, r) \times B_{e_1}(y, r),$$

so every D-open set is product-open.

(ii) Conversely, let $U = B_{d_1}(x, \alpha)$ and $V = B_{e_1}(y, \beta)$. For $r = \min\{\alpha, \beta\}/2$ we have

$$B_D((x, y), r) \subseteq U \times V,$$

so every basic product-open set is D-open.

Thus the two topologies coincide, and $X \times Y$ is metrizable.

Finite products. If (X_i, d_i) for $i = 1, \ldots, n$ are metric,

define on $X = \prod_{i=1}^n X_i$ the metric

$$D\Big((x_i), (x_i')\Big) = \sum_{i=1}^n \min\{1, d_i(x_i, x_i')\}.$$

Exactly the same containment argument shows that D induces the product topology on X. Hence any finite product of metrizable spaces is metrizable. □

Examples.

- $\mathbb{R}^n$ is metrizable (finite product of $(\mathbb{R}, |\cdot|)$).
- A finite product of discrete spaces is again discrete (hence metrizable).
- $S^1 \times [0, 1]$ is metrizable; for instance, with

  $$d\Big((\theta, t), (\phi, s)\Big) = \sqrt{d_{S^1}(\theta, \phi)^2 + |t - s|^2},$$

  where d_{S^1} is the arc-length distance on the circle.

Problem 10: A quotient that bridges two intervals

Statement. Let $X = [0, 1] \cup [2, 3]$ with the subspace topology of $\mathbb{R}$ and define $x \sim y$ iff $x = y$ or $\{x, y\} = \{1, 2\}$. Show that $X/\sim$ is homeomorphic to $[0, 2]$ with the usual topology.

Solution. Define $F : X \to [0,2]$ by

$$F(x) = \begin{cases} x, & x \in [0,1], \\ x-1, & x \in [2,3]. \end{cases}$$

Then F is continuous (it is continuous on each closed piece, and $F(1) = 1 = F(2)$), and it is constant on $\sim$-classes (the only nontrivial class is $\{1,2\}$).

Let $q : X \to X/\sim$ be the quotient map. By the universal property of the quotient, there exists a unique continuous $\tilde{F} : X/\sim \to [0,2]$ such that $\tilde{F} \circ q = F$.

We show $\tilde{F}$ is bijective. For $y \in [0,2]$,

$$F^{-1}(\{y\}) = \begin{cases} \{y\}, & 0 \le y < 1, \\ \{1,2\}, & y = 1, \\ \{y+1\}, & 1 < y \le 2, \end{cases}$$

so exactly one equivalence class maps to each y. Hence $\tilde{F}$ is bijective.

Since X is compact, the quotient $X/\sim$ is compact. The space $[0,2]$ is Hausdorff. Therefore a continuous bijection $\tilde{F} : X/\sim \to [0,2]$ is a homeomorphism. Thus $X/\sim \cong [0,2]$.

Remark. If we additionally identify $0 \sim 3$, the quotient becomes homeomorphic to S^1 (gluing the ends after bridging at $1 \sim 2$).

Why F is constant on classes. The $\sim$-classes are singletons $\{x\}$ for $x \neq 1,2$, and one special class $\{1,2\}$. For $x \neq 1,2$ we have a singleton class, so F is trivially constant there. For the special class,

$$F(1) = 1, \qquad F(2) = 2 - 1 = 1,$$

so F takes the same value on 1 and 2. Hence F is constant on every $\sim$-class.

Problem 11: Slices and a quotient of a product

Let X, Y be topological spaces and endow $X \times Y$ with the product topology.

(a) For each $y \in Y$, $X \times \{y\} \cong X$

Define

$$i_y : X \to X \times \{y\}, \quad i_y(x) = (x,y), \qquad p_y : X \times \{y\} \to X, \quad p_y(x,y) = x.$$

Then i_y is continuous (product of id_X with the constant map y), p_y is the restriction of the continuous projection π_X, and

$$p_y \circ i_y = \mathrm{id}_X, \qquad i_y \circ p_y = \mathrm{id}_{X\times\{y\}}.$$

Hence i_y is a homeomorphism $X \cong X \times \{y\}$.

(b) The quotient by vertical fibers is X (assuming $Y \neq \varnothing$)

Define $(x, y) \sim (x', y')$ iff $x = x'$. Let $q : X \times Y \to (X \times Y)/\!\sim$ be the quotient map and $\pi_X : X \times Y \to X$ the first projection. Since π_X is continuous and constant on $\sim$-classes, the universal property yields a unique continuous

$$\tilde{\pi} : (X \times Y)/\!\sim \longrightarrow X \quad \text{such that} \quad \tilde{\pi} \circ q = \pi_X.$$

Fix $y_0 \in Y$ and define $s : X \to (X \times Y)/\!\sim$ by $s(x) = q(x, y_0)$. Then s is continuous (it equals $q \circ i_{y_0}$).

For $x \in X$, $\tilde{\pi} \circ s(x) = \tilde{\pi}(q(x, y_0)) = \pi_X(x, y_0) = x$. For any $(x, y) \in X \times Y$ we have $(x, y) \sim (x, y_0)$, hence

$$s \circ \tilde{\pi}\Big(q(x, y)\Big) = s(x) = q(x, y_0) = q(x, y).$$

Thus $s = (\tilde{\pi})^{-1}$ and $\tilde{\pi}$ is a homeomorphism. Therefore $(X \times Y)/\!\sim \,\cong X$.

$$\begin{tikzcd}
X \times Y \arrow[d, "q"'] \arrow[dr, "\pi_X"] & \\
Q \arrow[r, "\tilde{\pi}"'] & X
\end{tikzcd}
\qquad
\begin{tikzcd}
X \arrow[r, "i_{y_0}"] \arrow[rr, bend right, "\mathrm{id}_X"'] & X \times Y \arrow[r, "q"] \arrow[dr, "\pi_X"'] & Q \arrow[loop right, "s\circ\tilde{\pi}"] \arrow[d, "\tilde{\pi}"] \\
& & X
\end{tikzcd}$$

Examples for (a).

- $X = \mathbb{R}$, $Y = [0, 1]$. For any $t \in [0, 1]$, the map $i_t : \mathbb{R} \to \mathbb{R} \times \{t\}$, $x \mapsto (x, t)$, is a homeomorphism with inverse $(x, t) \mapsto x$.
- $X = S^1$, $Y = \mathbb{R}$. Each slice $S^1 \times \{y\}$ is homeomorphic to S^1.
- $X = \{1, 2, 3\}$ (discrete), arbitrary Y. Each $\{1, 2, 3\} \times \{y\}$ is a 3–point discrete space $\cong X$.
- X arbitrary, Y non-Hausdorff (e.g. indiscrete). Still $X \times \{y\} \cong X$; the slice homeomorphism does not use separation axioms.

Examples for (b). Let $(x, y) \sim (x', y')$ iff $x = x'$ and assume $Y \neq \varnothing$.

- $X = \mathbb{R}$, $Y = [0,1]$. Each vertical segment $\{x\} \times [0,1]$ collapses to one point; the quotient is $\mathbb{R}$.
- $X = S^1$, $Y = \{0,1\}$. The two parallel circles collapse columnwise; the quotient is S^1.
- $X = \{1,2,3\}$ (discrete), $Y = \{a,b,c\}$. The 3×3 grid collapses each column to one point; the quotient is a 3–point discrete space $\cong X$.
- X arbitrary, Y indiscrete nonempty. Despite $X \times Y$ not being Hausdorff, the quotient by vertical fibers is homeomorphic to X via the factorization of π_X.

Why $Y \neq \varnothing$ is needed. If $Y = \varnothing$ then $X \times Y = \varnothing$ and $(X \times Y)/{\sim} = \varnothing \ncong X$ unless $X = \varnothing$.

Problem 12: $\mathbb{R}^2/\mathbb{Z}^2$ is a torus

Statement

Let $\sim$ on $\mathbb{R}^2$ be given by

$$(x,y) \sim (z,w) \quad \Longleftrightarrow \quad x - z \in \mathbb{Z} \;\text{ and }\; y - w \in \mathbb{Z}.$$

Show that $\mathbb{R}^2/{\sim}$ is homeomorphic to the embedded torus

$$T = \Big\{ \Big((2+\cos\theta)\cos\phi,\; (2+\cos\theta)\sin\phi,\; \sin\theta \Big) \;:\; \theta, \phi \in [0, 2\pi] \Big\} \subset \mathbb{R}^3.$$

Route A: via $S^1 \times S^1$

Step 1 (factor through the quotient). Define

$$F : \mathbb{R}^2 \longrightarrow S^1 \times S^1, \qquad F(x,y) = \Big(e^{2\pi i x},\, e^{2\pi i y} \Big).$$

Then:

- F is continuous.
- F is constant on $\sim$-classes (adding integers to x or y does not change the exponentials).

Hence F descends to a unique continuous

$$\tilde{F} :\; \mathbb{R}^2/{\sim} \;\longrightarrow\; S^1 \times S^1 \quad \text{with} \quad \tilde{F} \circ q = F \;\; (q \text{ the quotient map}).$$

Step 2 (bijectivity). Given $(u,v) \in S^1 \times S^1$, choose arguments $u = e^{2\pi i x}$, $v = e^{2\pi i y}$. Different choices differ by integers, hence determine the same class. Therefore $\tilde{F}$ is bijective.

Step 3 (homeomorphism). $\mathbb{R}^2/\!\sim$ is compact (homeomorphic to $[0,1]^2$ with opposite edges identified), and $S^1 \times S^1$ is Hausdorff. A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism; thus $\tilde{F}$ is a homeomorphism.

Step 4 (embed into $\mathbb{R}^3$). Define

$$\Phi : S^1 \times S^1 \to T, \qquad \Phi\Big(e^{i\theta}, e^{i\phi}\Big) = \Big((2+\cos\theta)\cos\phi,\ (2+\cos\theta)\sin\phi,\ \sin\theta\Big).$$

Φ is a homeomorphism onto T. Therefore

$$\mathbb{R}^2/\!\sim\ \cong\ S^1 \times S^1\ \cong\ T.$$

Route B: direct map to the embedded torus

Step 1 (define a $\mathbb{Z}^2$-periodic map). Set

$$G : \mathbb{R}^2 \to T, \qquad G(x,y) = \Big((2+\cos(2\pi x))\cos(2\pi y),\ (2+\cos(2\pi x))\sin(2\pi y),\ \sin(2\pi x)\Big).$$

Then G is continuous and satisfies $G(x+m, y+n) = G(x,y)$ for all $m,n \in \mathbb{Z}$, so G is constant on $\sim$-classes and factors through a continuous

$$\tilde{G} :\ \mathbb{R}^2/\!\sim\ \longrightarrow\ T.$$

Step 2 (bijective). For any $(\theta,\phi) \in [0,2\pi]^2$, the point of T with those angles equals

$$G\left(\frac{\theta}{2\pi}, \frac{\phi}{2\pi}\right).$$

Changing (θ,ϕ) by integer multiples of 2π changes (x,y) by integers, i.e. stays in the same class. Thus $\tilde{G}$ is bijective.

Step 3 (homeomorphism). As above, $\mathbb{R}^2/\!\sim$ is compact and T is Hausdorff (subspace of $\mathbb{R}^3$), hence the continuous bijection $\tilde{G}$ is a homeomorphism.

Variants

- $\mathbb{R}/\mathbb{Z} \cong S^1$ (the 1-torus).
- $\mathbb{R}^n/\mathbb{Z}^n \cong (S^1)^n$ (the n-torus).
- Identifying only the y-coordinate modulo $\mathbb{Z}$ gives the cylinder $\mathbb{R} \times S^1$.

Description of the map $G : \mathbb{R}^2 \to T$. Define

$$G(x, y) = \Big((2 + \cos(2\pi x)) \cos(2\pi y),\ (2 + \cos(2\pi x)) \sin(2\pi y),\ \sin(2\pi x)\Big).$$

Interpretation via angles. Set

$$\theta = 2\pi x, \qquad \phi = 2\pi y.$$

Then

$$G(x, y) = \big((2 + \cos\theta) \cos\phi,\ (2 + \cos\theta) \sin\phi,\ \sin\theta\big),$$

which is the standard parametrization of the embedded torus $T \subset \mathbb{R}^3$ with major radius 2 and minor radius 1. Here θ moves around the small circular cross–section (the "tube"), producing the vertical coordinate $z = \sin\theta$ and the tube radius 1, while ϕ rotates the cross–section around the z–axis; the distance from the z–axis is $2 + \cos\theta$ and multiplying by $(\cos\phi, \sin\phi)$ sweeps the big circle.

Periodicity and factorization through the quotient. For every $(m, n) \in \mathbb{Z}^2$,

$$G(x + m, y + n) = G(x, y),$$

since only $2\pi x$ and $2\pi y$ appear in trigonometric functions. Thus G is constant on cosets of $\mathbb{Z}^2$ in $\mathbb{R}^2$, i.e. on the equivalence classes for the relation $(x, y) \sim (x', y') \iff (x - x', y - y') \in \mathbb{Z}^2$. Let $q : \mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2$ be the quotient map. By the universal property of the quotient, there exists a unique continuous

$$\tilde{G} :\ \mathbb{R}^2/\mathbb{Z}^2 \longrightarrow T \quad \text{such that} \quad \tilde{G} \circ q = G.$$

Surjectivity and injectivity on the quotient. Given any point of T written as

$$\big((2 + \cos\theta) \cos\phi,\ (2 + \cos\theta) \sin\phi,\ \sin\theta\big) \quad (\theta, \phi \in [0, 2\pi]),$$

choose $(x, y) = (\theta/2\pi,\ \phi/2\pi) \in \mathbb{R}^2$; then $G(x, y)$ equals that point. Different choices of (θ, ϕ) differ by integer multiples of 2π and hence change (x, y) by integers only; therefore they lie in the same class in $\mathbb{R}^2/\mathbb{Z}^2$. Consequently $\tilde{G}$ is bijective.

Homeomorphism. The space $\mathbb{R}^2/\mathbb{Z}^2$ is compact (homeomorphic to the unit square $[0,1]^2$ with opposite edges identified), and $T \subset \mathbb{R}^3$ is Hausdorff. Hence the continuous bijection $\tilde{G} : \mathbb{R}^2/\mathbb{Z}^2 \to T$ is a homeomorphism.

Why the torus T? The relation $(x, y) \sim (x + m, y + n)$, $m, n \in \mathbb{Z}$, identifies opposite edges of the unit square $[0,1]^2$, producing the compact surface $[0,1]^2/\sim$ which is homeomorphic to $S^1 \times S^1$, i.e. a (topological) torus. The set

$$T = \{((2 + \cos\theta)\cos\phi, (2 + \cos\theta)\sin\phi, \sin\theta) : \theta, \phi \in [0, 2\pi]\} \subset \mathbb{R}^3$$

is a standard embedded model of this torus. The map

$$G(x, y) = \Big((2 + \cos 2\pi x)\cos 2\pi y,\ (2 + \cos 2\pi x)\sin 2\pi y,\ \sin 2\pi x\Big)$$

is $\mathbb{Z}^2$–periodic and hence factors through the quotient, yielding a homeomorphism $\tilde{G} : \mathbb{R}^2/\mathbb{Z}^2 \to T$.

Problem 13: The line with two origins

Let

$$X = \{(x, y) \in \mathbb{R}^2 : y = \pm 1\}$$

with the subspace topology from $\mathbb{R}^2$, and define an equivalence relation on X by

$$(x, y) \sim (x', y') \iff x' = x \neq 0 \quad\text{and}\quad y' = -y.$$

Let $Q = X/\sim$ and $q : X \to Q$ be the quotient map. Denote

$$o_+ = q(0, 1), \qquad o_- = q(0, -1).$$

(a) Every point of Q has a neighborhood homeomorphic to $\mathbb{R}$

Case 1: points coming from $x \neq 0$. Fix $a \neq 0$ and choose $\varepsilon > 0$ so that $(a - \varepsilon, a + \varepsilon)$ does not meet 0. Then

$$U = q\Big((a - \varepsilon, a + \varepsilon) \times \{1\}\Big) = q\Big((a - \varepsilon, a + \varepsilon) \times \{-1\}\Big)$$

and the map $t \mapsto q(t, 1)$ is a homeomorphism $(a - \varepsilon, a + \varepsilon) \xrightarrow{\cong} U$.

Case 2: the point o_+. For small $\varepsilon > 0$, define

$$\phi_+ : (-\varepsilon, \varepsilon) \longrightarrow Q, \qquad \phi_+(t) = \begin{cases} q(t, 1), & t \leq 0, \\ q(t, -1), & t > 0. \end{cases}$$

Then ϕ_+ is continuous, bijective onto $U_+ = \phi_+((-\varepsilon, \varepsilon))$, and its inverse is continuous (it pulls back opens to unions of opens in the two lines). Hence $U_+ \cong \mathbb{R}$.

Case 3: the point o_-. Define

$$\phi_- : (-\varepsilon, \varepsilon) \to Q, \qquad \phi_-(t) = \begin{cases} q(t,-1), & t \le 0, \\ q(t,1), & t > 0. \end{cases}$$

Then $U_- = \phi_-((-\varepsilon, \varepsilon))$ is a neighborhood of o_- homeomorphic to $\mathbb{R}$.

Thus each point of Q has a neighborhood homeomorphic to $\mathbb{R}$.

(b) Q is not Hausdorff

Let $U \ni o_+$ and $V \ni o_-$ be arbitrary open neighborhoods. By the constructions above, for all sufficiently small $t > 0$ we have $q(t,1) = q(t,-1) \in U$ and also $q(t,1) = q(t,-1) \in V$. Hence $U \cap V \neq \varnothing$, so o_+ and o_- cannot be separated by disjoint open sets. Therefore Q is not Hausdorff.

Remark. Q is the classical “line with two origins”: it is locally homeomorphic to $\mathbb{R}$, but it fails to be Hausdorff.

Quotient equivalence classes in our examples.

(10) $X = [0,1] \cup [2,3]$ with $1 \sim 2$.

- Classes: $\{x\}$ for every $x \in X \setminus \{1,2\}$, and the special class $\{1,2\}$.

(11b) $X \times Y$ with $(x,y) \sim (x',y') \iff x = x'$ (assume $Y \neq \varnothing$).

- Classes: for each $x \in X$, the vertical fiber $\{x\} \times Y$.

(12) $\mathbb{R}^2$ with $(x,y) \sim (x',y') \iff (x-x', y-y') \in \mathbb{Z}^2$.

- Classes: lattice cosets $(x,y) + \mathbb{Z}^2 = \{(x+m, y+n) : m,n \in \mathbb{Z}\}$.
- Equivalently: one representative in $[0,1) \times [0,1)$, with opposite edges identified.

(13) $X = \{(x,y) \in \mathbb{R}^2 : y = \pm 1\}$ with $(x,y) \sim (x',y') \iff x = x' \neq 0,\ y' = -y$.

- Classes: for $x \neq 0$, the pair $\{(x,1),(x,-1)\}$; for $x = 0$, the singletons $\{(0,1)\}$ and $\{(0,-1)\}$.

Problem 14: Compactness of Cocountable vs. Cofinite Topologies

Definitions. On a set X, let

$\tau = \{\varnothing\} \cup \{\, U \subseteq X :\ X \setminus U \text{ is countable} \,\}$ and $\rho = \{\varnothing\} \cup \{\, U \subseteq X :\ X \setminus U \text{ is finite} \,\}$.

(A) Cocountable topology (X, τ)

- If X is finite, then every subset is open (discrete topology), hence (X, τ) is compact.
- If X is infinite, then (X, τ) is *not* compact. Choose a countably infinite subset $A = \{a_1, a_2, \dots\} \subseteq X$. For $n \geq 1$ set $C_n = \{a_n, a_{n+1}, \dots\}$ (countable) and $U_n = X \setminus C_n \in \tau$. Then $\bigcup_{n \geq 1} U_n = X$ since $\bigcap_{n \geq 1} C_n = \varnothing$, but for each k,

  $$\bigcup_{n=1}^{k} U_n = X \setminus \Big(\bigcap_{n=1}^{k} C_n \Big) = X \setminus \{a_k, a_{k+1}, \dots\} \neq X,$$

  so no finite subcover exists. Hence (X, τ) is compact iff X is finite (the empty space is compact as usual).

(B) Cofinite topology (X, ρ)

(X, ρ) is compact for *every* X. Indeed, given an open cover $\{U_\alpha\}_{\alpha \in A}$ of X, pick a nonempty U_{α_0}. Then $F = X \setminus U_{\alpha_0}$ is finite. For each $x \in F$ choose $U_{\alpha(x)}$ with $x \in U_{\alpha(x)}$. Thus

$$X = U_{\alpha_0} \;\cup\; \bigcup_{x \in F} U_{\alpha(x)}$$

is a finite subcover.

Examples. On $X = \mathbb{R}$ the cocountable topology is not compact (use the $U_n = \mathbb{R} \setminus \{a_n, a_{n+1}, \dots\}$ cover for a countably infinite subset $\{a_n\}$), while the cofinite topology is compact. If X is finite, both topologies coincide with the discrete topology and are compact.

Problem 15: Compact Hausdorff spaces separate closed sets

Statement. Let X be compact Hausdorff. If $F_1, F_2 \subset X$ are closed and disjoint, then there exist disjoint open sets $G_1, G_2 \subset X$ with $F_1 \subset G_1$ and $F_2 \subset G_2$.

Proof (compact Hausdorff $\Rightarrow$ normal). Since F_1, F_2 are closed in a compact space, they are compact.

Step 1. Fix $x \in F_1$. For each $y \in F_2$ choose disjoint open sets $U_{x,y} \ni x$ and $V_{x,y} \ni y$ (Hausdorff). The family $\{V_{x,y} : y \in F_2\}$ covers F_2, so compactness of F_2 yields $y_1, \dots, y_n$ with $F_2 \subset \bigcup_{i=1}^{n} V_{x,y_i}$. Set

$$U_x = \bigcap_{i=1}^{n} U_{x,y_i}, \qquad V_x = \bigcup_{i=1}^{n} V_{x,y_i}.$$

Then U_x, V_x are open, $x \in U_x$, $F_2 \subset V_x$, and $U_x \cap V_x = \varnothing$.

Step 2. The family $\{U_x : x \in F_1\}$ covers F_1. By compactness of F_1 there exist $x_1, \ldots, x_m \in F_1$ with $F_1 \subset \bigcup_{j=1}^m U_{x_j}$. Define

$$G_1 = \bigcup_{j=1}^{m} U_{x_j}, \qquad G_2 = \bigcap_{j=1}^{m} V_{x_j}.$$

Then G_1, G_2 are open, $F_1 \subset G_1$, $F_2 \subset G_2$, and

$$G_1 \cap G_2 \subset \bigcup_{j=1}^{m} (U_{x_j} \cap V_{x_j}) = \varnothing,$$

so G_1 and G_2 are disjoint. □

Remarks. Compact Hausdorff $\Rightarrow$ normal. Without compactness, Hausdorff does not imply normal: the Sorgenfrey plane $\mathbb{S} \times \mathbb{S}$ is Hausdorff but not normal. Without Hausdorffness, even compact spaces (e.g. the cofinite topology on an infinite set) fail to separate points and sets.

Normal, Tychonoff ($T_{3\frac{1}{2}}$), and Sorgenfrey—examples and counterexamples

Separation axioms (recap). T_1: singletons are closed; T_2 (Hausdorff): points have disjoint nbhds;

T_3 (regular): T_1 + point vs. closed set separated by disjoint opens;

$T_{3\frac{1}{2}}$ (Tychonoff): T_1 + point vs. closed set separated by a continuous real-valued function (completely regular Hausdorff);

T_4 (normal): T_1 + disjoint closed sets separated by disjoint opens;

T_5 (completely normal): every two separated sets have disjoint open nbhds (equivalently: hereditarily normal);

T_6 (perfectly normal): normal and every closed set is a G_δ.

Examples

- **Metric spaces** (e.g. $\mathbb{R}^n$, compact metric spaces): perfectly normal (T_6) $\Rightarrow$ completely normal (T_5) $\Rightarrow$ normal (T_4) $\Rightarrow$ Tychonoff.
- **Compact Hausdorff spaces**: normal (by the standard theorem).

- **Finite products of metric spaces**: metric $\Rightarrow$ normal.
- **Tychonoff products**: Any product of Tychonoff spaces is Tychonoff (Tychonoff theorem).
- **Sorgenfrey line** $\mathbb{S}$ (base $[a, b)$): T_6 (perfectly normal), first countable, separable; not second countable, not Lindelöf.

Counterexamples / Cautions

- **Sorgenfrey plane** $\mathbb{S} \times \mathbb{S}$: Tychonoff but *not normal*. Hence product of normal spaces may fail to be normal.
- **Niemytzki (Moore) plane**: T_1 and completely regular (so Tychonoff) but not normal.
- **Cofinite/cocountable topologies on an infinite set**: T_1 but not Hausdorff; thus not T_3, not Tychonoff, not normal.

Takeaways.

- Normal $\Rightarrow$ Tychonoff is false; Tychonoff $\not\Rightarrow$ normal (witness $\mathbb{S} \times \mathbb{S}$).
- Compact Hausdorff $\Rightarrow$ normal; metric $\Rightarrow$ perfectly normal.
- Products preserve Tychonoff, but not normality in general.

Problem. Recall from lectures that a non-empty topological space (X, τ) is said to be connected if there is no separation of X by open, non-empty, disjoint subsets. Show that (X, τ) is connected if and only if the only subsets of X that are open and closed in X are $\varnothing$ and X itself.

Definitions. A *separation* of X is a pair of sets $U, V \subseteq X$ such that $U \neq \varnothing$, $V \neq \varnothing$, $U \cap V = \varnothing$, each of U and V is open, and $X = U \cup V$. The space X is *connected* if there is no separation of X. A set $A \subseteq X$ is *clopen* if it is both open and closed (*closed* meaning $X \setminus A$ is open).

Key observations.

1. If $X = U \cup V$ is a separation, then U is clopen: U is open by assumption and $X \setminus U = V$ is also open, hence U is closed. Similarly for V.
2. If $A \subsetneq X$ is non-empty and clopen, then A and $X \setminus A$ are disjoint, non-empty, open sets whose union is X, i.e., they form a separation of X.

Proof. *($\Rightarrow$)* Suppose X is connected. Let $A \subseteq X$ be clopen. If $A \notin \{\varnothing, X\}$, then by Observation 2 the pair $(A, X \setminus A)$ is a separation of X, contradicting connectedness. Hence the only clopen subsets are $\varnothing$ and X.

($\Leftarrow$) Suppose the only clopen subsets of X are $\varnothing$ and X. If X were disconnected, there would exist a separation $X = U \cup V$ with U, V non-empty, open, disjoint. Then U is closed since $X \setminus U = V$ is open, so U is a non-trivial clopen subset of X, contradicting the assumption. Therefore X is connected.

Conclusion. A topological space (X, τ) is connected if and only if the only clopen subsets of X are $\varnothing$ and X itself.

Problem. Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$, with canonical metric $d(x, y) = \|x - y\|$. Show that the open and closed balls of such (X, d) are path connected.

Definitions. For $x_0 \in X$ and $r > 0$,

$$B(x_0, r) = \{x \in X : \|x - x_0\| < r\}, \qquad \overline{B}(x_0, r) = \{x \in X : \|x - x_0\| \leq r\}.$$

A subset $A \subset X$ is *path connected* if for any $x, y \in A$ there exists a continuous map $\gamma : [0, 1] \to A$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Lemma (Convex $\Rightarrow$ path connected). If $A \subset X$ is convex, then for any $x, y \in A$ the map $\gamma(t) = (1 - t)x + ty$ ($t \in [0, 1]$) is a continuous path in A from x to y. Hence A is path connected.

Claim. Open and closed balls are convex.

Proof. Let x, y lie in $\overline{B}(x_0, r)$ and $t \in [0, 1]$. Then, by the triangle inequality and absolute homogeneity,

$$\|(1-t)x+ty-x_0\| = \|(1-t)(x-x_0)+t(y-x_0)\| \leq (1-t)\|x-x_0\|+t\|y-x_0\| \leq (1-t)r+tr = r.$$

Thus $(1 - t)x + ty \in \overline{B}(x_0, r)$, proving convexity of the closed ball. If $x, y \in B(x_0, r)$, the same estimate with strict inequalities gives

$$\|(1 - t)x + ty - x_0\| < (1 - t)r + tr = r,$$

so $(1 - t)x + ty \in B(x_0, r)$, proving convexity of the open ball. $\square$

Proof of the statement. By the lemma, any convex subset of a normed space is path connected using the straight-line path $\gamma(t) = (1 - t)x + ty$. Since both $B(x_0, r)$ and $\overline{B}(x_0, r)$ are convex, they are path connected. $\square$

Problem 3 — Path-connected $\Rightarrow$ Connected. Show that if (X, τ) is path-connected, then it is also connected.

Definitions. A space X is *path-connected* if for every $x, y \in X$ there exists a continuous map $\gamma : [0, 1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. A space is *connected* if it admits no separation $X = U \cup V$ with U, V nonempty, disjoint, open sets.

Facts. (1) The interval $[0,1]$ is connected. (2) Continuous images of connected sets are connected. (3) A connected subset of a space cannot meet two disjoint nonempty open sets that separate the space.

Proof. Assume X is path-connected. Suppose, to the contrary, that there is a separation $X = U \cup V$ with U, V nonempty, open, and disjoint. Choose $x \in U$ and $y \in V$. By path-connectedness there exists a continuous $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. Then $\gamma([0,1])$ is connected as the continuous image of the connected set $[0,1]$. However, $\gamma([0,1]) \subset U \cup V$ and meets both U and V (since $\gamma(0) = x \in U$ and $\gamma(1) = y \in V$), which contradicts the fact that a connected set cannot be expressed as the union of two disjoint nonempty open subsets. Therefore no such separation exists, and X is connected.

Alternate preimage proof. The sets $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are disjoint, nonempty, open in $[0,1]$, and cover $[0,1]$. This contradicts the connectedness of $[0,1]$. Hence X is connected.

Examples.

- Any interval in $\mathbb{R}$ is path-connected (via straight-line paths), hence connected.
- Open or closed balls in a normed vector space are convex, thus path-connected; therefore connected.
- $\mathbb{R}^n \setminus \{0\}$ is path-connected for $n \geq 2$ (join points by radial segments and an arc), hence connected.

Counterexample to the converse. The topologist's sine curve $S = \{(x, \sin(1/x)) : x \in (0,1]\} \cup (\{0\} \times [-1,1]) \subset \mathbb{R}^2$ is connected but not path-connected.

Problem 4 — Path-components form an equivalence. Show that the relation $\sim$ on X defined by $x \sim y$ iff there exists a continuous path $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$ is an equivalence relation. (The equivalence classes are the path-components of X.)

Tools. For paths $\gamma, \eta : [0,1] \to X$:

- Constant path at x: $c_x(t) \equiv x$.
- Reversal: $\overline{\gamma}(t) = \gamma(1-t)$ (joins $\gamma(1)$ to $\gamma(0)$).
- Concatenation:
$$(\gamma * \eta)(t) = \begin{cases} \gamma(2t), & t \in [0, \frac{1}{2}], \\ \eta(2t-1), & t \in [\frac{1}{2}, 1], \end{cases}$$
a continuous path that joins $\gamma(0)$ to $\eta(1)$ provided $\gamma(1) = \eta(0)$.

Proof. We verify the three axioms.

- *Reflexive.* For any $x \in X$, c_x is a path from x to x, so $x \sim x$.

- *Symmetric.* If $x \sim y$, let γ be a path from x to y. Then $\overline{\gamma}(t) = \gamma(1-t)$ is a path from y to x, hence $y \sim x$.
- *Transitive.* If $x \sim y$ via γ and $y \sim z$ via η, the concatenation $\gamma * \eta$ is a path from x to z, thus $x \sim z$.

Therefore $\sim$ is an equivalence relation. Its equivalence classes are the path-components of X, i.e., maximal path-connected subsets that partition X. $\square$

Examples.

1. $\mathbb{R}$: one path-component (the whole space).
2. $\mathbb{R} \setminus \{0\}$: two path-components $(-\infty, 0)$ and $(0, \infty)$.
3. $\mathbb{R}^2 \setminus \{0\}$: one path-component (you can avoid the origin).
4. Discrete topology: every singleton $\{x\}$ is a path-component.
5. Indiscrete topology (nonempty X): X itself is the unique path-component.
6. Topologist's sine curve $S = \{(x, \sin(1/x)) : x \in (0,1]\} \cup (\{0\} \times [-1,1])$: the subset $\{(x, \sin(1/x)) : x \in (0,1]\}$ is one path-component, and each point on the vertical segment $\{0\} \times [-1,1]$ is a singleton path-component.

Problem 5 — Basic rules of multivariable differentiation

Statement. Verify the following for differentiable maps between Euclidean spaces.

(a) *(Sum and scalar multiplication)* If $f, g : \mathbb{R}^n \to \mathbb{R}^m$ are differentiable at x_0 and $\lambda \in \mathbb{R}$, then $\lambda f + g$ is differentiable at x_0 with

$$D(\lambda f + g)\Big|_{x_0} = \lambda\, Df\Big|_{x_0} + Dg\Big|_{x_0}.$$

(b) *(Product rule for the inner product)* If $f, g : \mathbb{R}^n \to \mathbb{R}^k$ are differentiable at x_0, then $F(x) := \langle f(x), g(x) \rangle$ is differentiable at x_0 and

$$DF\Big|_{x_0}(h) = \langle Df|_{x_0}(h),\, g(x_0)\rangle + \langle f(x_0),\, Dg|_{x_0}(h)\rangle.$$

(c) *(Chain rule)* If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at x_0 and $g : \mathbb{R}^m \to \mathbb{R}^d$ is differentiable at $f(x_0)$, then $g \circ f$ is differentiable at x_0 and

$$D(g \circ f)\Big|_{x_0}(h) = Dg\Big|_{f(x_0)}\Big(Df\Big|_{x_0}(h)\Big).$$

Theory (Fréchet derivative). A map $F : \mathbb{R}^n \to \mathbb{R}^p$ is differentiable at x_0 iff there exists a linear $L : \mathbb{R}^n \to \mathbb{R}^p$ with

$$F(x_0 + h) = F(x_0) + L(h) + o(\|h\|) \quad (h \to 0),$$

and then $L = DF|_{x_0}$. Continuous bilinear maps (e.g. the inner product) satisfy

$$B(u+u', v+v') = B(u,v) + B(u',v) + B(u,v') + O(\|u'\|\,\|v'\|).$$

Proofs.

(a) Sum/scalar multiple. Write

$$f(x_0+h) = f(x_0) + Df|_{x_0}(h) + r_f(h), \qquad g(x_0+h) = g(x_0) + Dg|_{x_0}(h) + r_g(h),$$

with $r_f(h), r_g(h) = o(\|h\|)$. Then

$$(\lambda f + g)(x_0+h) = (\lambda f+g)(x_0) + \Big(\lambda Df|_{x_0} + Dg|_{x_0}\Big)(h) + o(\|h\|),$$

so $D(\lambda f + g)|_{x_0} = \lambda Df|_{x_0} + Dg|_{x_0}$.

(b) Product rule. Let $F(x) = \langle f(x), g(x)\rangle$. Using the expansions above and bilinearity,

$$\begin{aligned} F(x_0+h) &= \Big\langle f(x_0) + Df|_{x_0}(h) + r_f(h),\ g(x_0) + Dg|_{x_0}(h) + r_g(h)\Big\rangle \\ &= F(x_0) + \langle Df|_{x_0}(h),\, g(x_0)\rangle + \langle f(x_0),\, Dg|_{x_0}(h)\rangle + o(\|h\|). \end{aligned}$$

(c) Chain rule. Let $y_0 = f(x_0)$ and write

$$f(x_0+h) = y_0 + Df|_{x_0}(h) + r(h), \qquad r(h) = o(\|h\|).$$

Since g is differentiable at y_0,

$$g(y_0+k) = g(y_0) + Dg|_{y_0}(k) + \rho(k), \qquad \rho(k) = o(\|k\|).$$

Apply with $k = Df|_{x_0}(h) + r(h)$:

$$\begin{aligned} (g\circ f)(x_0+h) &= g(y_0 + Df|_{x_0}(h) + r(h)) \\ &= (g\circ f)(x_0) + \Big(Dg|_{y_0}\circ Df|_{x_0}\Big)(h) + Dg|_{y_0}\Big(r(h)\Big) + \rho(Df|_{x_0}(h) + r(h))\,, \end{aligned}$$

and the last two terms are $o(\|h\|)$. Hence

$$D(g\circ f)\Big|_{x_0} = Dg\Big|_{f(x_0)} \circ Df\Big|_{x_0}.$$

$\square$

Quick examples.

(a) $f(x) = Ax, \quad g(x) = b$ (constant), $\lambda \in \mathbb{R}$.
$Df\Big|_{x_0} = A, \quad Dg\Big|_{x_0} = 0$, so

$$D(\lambda f + g)\Big|_{x_0} = \lambda A.$$

(b) $f(x) = (x_1, x_2), \quad g(x) = (x_1^2, x_2^2)$ on $\mathbb{R}^2$.

$$Df\Big|_{x_0} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad Dg\Big|_{x_0} = \begin{pmatrix} 2x_{01} & 0 \\ 0 & 2x_{02} \end{pmatrix}.$$

Then, by the inner–product product rule, for $h = (h_1, h_2)$,

$$\begin{aligned} D(f \cdot g)\Big|_{x_0}(h) &= \langle Df|_{x_0}(h),\, g(x_0)\rangle + \langle f(x_0),\, Dg|_{x_0}(h)\rangle \\ &= \langle (h_1, h_2),\, (x_{01}^2, x_{02}^2)\rangle + \langle (x_{01}, x_{02}),\, (2x_{01}h_1, 2x_{02}h_2)\rangle \\ &= 3x_{01}^2\, h_1 + 3x_{02}^2\, h_2. \end{aligned}$$

(c) $f(x_1, x_2) = \big(x_1^2 + x_2,\ \sin x_1\big), \quad g(y_1, y_2) = y_1 y_2.$

$$Df\Big|_{x_0} = \begin{pmatrix} 2x_{01} & 1 \\ \cos x_{01} & 0 \end{pmatrix}, \qquad Dg\Big|_{f(x_0)} = \begin{pmatrix} y_2 & y_1 \end{pmatrix} \text{ evaluated at } y = f(x_0) = (x_{01}^2 + x_{02},\, \sin x_{01}).$$

Hence

$$D(g \circ f)\Big|_{x_0} = Dg\Big|_{f(x_0)} Df\Big|_{x_0} \quad \text{(matrix product)}.$$

Fréchet derivative — full picture

Setup. Let E, F be normed (usually Banach) spaces over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, $U \subset E$ open, $f : U \to F$, $x_0 \in U$.

Definition (Fréchet differentiable at x_0). There exists a bounded linear map $L : E \to F$ such that

$$\lim_{h \to 0} \frac{\|\, f(x_0 + h) - f(x_0) - Lh \,\|_F}{\|h\|_E} = 0.$$

Then L is the Fréchet derivative and is unique; write $Df(x_0)$ or $Df\Big|_{x_0}$.

Interpretation.

$$f(x_0 + h) = f(x_0) + Df(x_0)h + o(\|h\|),$$

i.e. a best linear approximation with error strictly smaller than linear in $\|h\|$.

Equivalent characterizations.

- *Remainder form:* with $r(h) = f(x_0 + h) - f(x_0) - Df(x_0)h$ we have $\|r(h)\| = o(\|h\|)$.
- *Directional uniformity:* if f is Fréchet differentiable, then for every unit u

$$\lim_{t \to 0} \frac{f(x_0 + tu) - f(x_0)}{t} = Df(x_0)u,$$

and the convergence is uniform in u on compact subsets of the unit sphere.

Relation to Gâteaux derivative. The Gâteaux derivative at x_0 is

$$D_G f(x_0; h) = \lim_{t \to 0} \frac{f(x_0 + th) - f(x_0)}{t}$$

(if it exists for each h). If f is Fréchet differentiable then $D_G f(x_0; h) = Df(x_0)h$. The converse need not hold; however, $D_G f$ being linear/continuous in h with mild uniformity (e.g. local Lipschitz) often upgrades to Fréchet. In finite dimensions, "all partials exist and are continuous" $\Rightarrow$ Fréchet (C^1).

Finite-dimensional form. For $f : \mathbb{R}^n \to \mathbb{R}^m$, $Df(x_0)$ is the Jacobian $J_f(x_0)$; the linear map is $h \mapsto J_f(x_0)h$. All norms on finite-dimensional spaces are equivalent, so the definition is norm-independent.

Key properties.

- *Linearity:* $D(\alpha f + \beta g) = \alpha\, Df + \beta\, Dg$.
- *Product rules:* for a continuous bilinear B,

$$D\Big(x \mapsto B(f, g)\Big) \;=\; \Big(h \mapsto B(Df(\,\cdot\,)h,\, g)\Big) + \Big(h \mapsto B(f,\, Dg(\,\cdot\,)h)\Big).$$

  (Covers inner products, matrix products, etc.)
- *Chain rule:* if $f : U \to F$ is Fréchet at x_0 and $g : V \to G$ is Fréchet at $f(x_0)$, then $g \circ f$ is Fréchet at x_0 and

$$D(g \circ f)(x_0) = Dg\Big(f(x_0)\Big) \circ Df(x_0).$$

- *Continuity:* Fréchet differentiability implies continuity at x_0.

Holomorphic maps and Fréchet differentiability.

One complex variable ($\mathbb{C} \to \mathbb{C}$). A function $F : \mathbb{C} \to \mathbb{C}$ is holomorphic at z_0 iff the *complex* Fréchet derivative exists at z_0, i.e. there is a $\mathbb{C}$-linear map $L(w) = c\,w$ with

$$F(z_0 + h) = F(z_0) + c\,h + o(|h|).$$

Complex differentiability on an open set already implies C^∞, real-Fréchet differentiability of all orders, the Cauchy–Riemann equations, and a convergent power series. (Complex differentiability is strictly stronger than real differentiability.)

Caution. Real Fréchet differentiability + CR at a point $\Rightarrow$ complex differentiable *at that point*; mere directional (Gâteaux) complex differentiability does *not* suffice.

Several complex variables / complex Banach spaces. For complex Banach spaces E, F, a map $F : U \subset E \to F$ is holomorphic iff it is complex Fréchet differentiable at each point (i.e. admits a bounded $\mathbb{C}$-linear derivative $DF(x)$). Classically: *locally bounded + complex Gâteaux differentiable* $\Rightarrow$ holomorphic $\Rightarrow$ complex Fréchet differentiable (derivative automatically continuous and $\mathbb{C}$-linear). For $F : \mathbb{C}^n \to \mathbb{C}^m$, holomorphic $\iff$ each component is holomorphic $\iff$ F is complex Fréchet differentiable and the real Jacobian satisfies the block Cauchy–Riemann relations.

Real vs complex derivative. Identify $\mathbb{C}^n \simeq \mathbb{R}^{2n}$. If F is holomorphic, its real Fréchet derivative $D_{\mathbb{R}}F$ is $\mathbb{R}$-linear and commutes with the complex structure J (i.e. satisfies Cauchy–Riemann). Conversely, real Fréchet differentiable with CR plus mild regularity $\Rightarrow$ complex Fréchet derivative exists and equals the complex one.

Tiny examples.

- *Real Fréchet (finite-dim).* $f(x,y) = (x^2 + xy,\ e^x \cos y)$. Then

$$Df(x,y) = \begin{pmatrix} 2x+y & x \\ e^x \cos y & -e^x \sin y \end{pmatrix}.$$

- *Complex holomorphic $\Leftrightarrow$ complex Fréchet.* $F(z) = z^3$.

$$DF(z)(h) = 3z^2\, h \quad (\mathbb{C}\text{-linear}).$$

  As a real map $\mathbb{R}^2 \to \mathbb{R}^2$, the Jacobian satisfies CR.

Gâteaux but not Fréchet — two examples.

(A) Discontinuous at the point.

$$f(x,y) = \begin{cases} \dfrac{x^2\, y}{x^4 + y^2}, & (x,y) \neq (0,0), \\ 0, & (0,0). \end{cases}$$

For any $h = (a,b)$,

$$\frac{f(th) - f(0)}{t} = \frac{t\, a^2 b}{t^2 a^4 + b^2} \to 0 \quad (t \to 0),$$

so the Gâteaux derivative at $(0,0)$ is 0. Yet along $y = x^2$, $f(x,x^2) = \frac{1}{2} \not\to 0$, hence f is not continuous and therefore not Fréchet at $(0,0)$.

(B) Continuous but still not Fréchet.

$$g(x,y) = \begin{cases} \dfrac{x^3}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (0,0). \end{cases}$$

For any $h = (a,b)$,

$$\frac{g(th) - g(0)}{t} = \frac{t\, a^3}{a^2 + b^2} \to 0,$$

so the Gâteaux derivative at $(0,0)$ is 0. Also $|g(x,y)| \le |x| \to 0$, so g is continuous. However,

$$\frac{|g(t,t^2)|}{\sqrt{t^2+t^4}} = \frac{\left|\frac{t}{1+t^2}\right|}{|t|\sqrt{1+t^2}} = \frac{1}{(1+t^2)^{3/2}} \xrightarrow[t\to 0]{} 1 \neq 0,$$

so the Fréchet condition with linear part 0 fails; g is not Fréchet differentiable at $(0,0)$.

Problem 6 — Connectedness and Path-connectedness in $\mathbb{R}^2$

Problem. Which of the following subsets of $\mathbb{R}^2$ with the Euclidean topology are connected? Which are path-connected? (And why?)

(a) $\{(x,y) \in \mathbb{R}^2 : \|(x,y)-(-1,0)\| \le 1 \text{ or } \|(x,y)-(1,0)\| < 1\}$

(b) $\{(x,y) \in \mathbb{R}^2 :\ x = 0 \text{ or } y = qx \text{ for } q \in \mathbb{Q}\}$

(c) $\{(x,y) \in \mathbb{R}^2 :\ x = 0 \text{ or } y = qx \text{ for } q \in \mathbb{Q}\} \setminus \{(0,0)\}$

(d) $S \subset \mathbb{R}^2$ any star-shaped domain, i.e. there exists $x_0 \in S$ such that for all $x \in S$ the line segment $[x_0, x] \subset S$.

Theory. A subset $A \subset \mathbb{R}^2$ is *connected* if it cannot be written as the disjoint union of two non-empty open subsets in the subspace topology. It is *path-connected* if for every $x, y \in A$ there exists a continuous $\gamma : [0,1] \to A$ with $\gamma(0) = x$, $\gamma(1) = y$. In $\mathbb{R}^n$, path-connected $\Rightarrow$ connected, but not conversely.

Solutions.

(a) Two tangent disks. Let $D_1 = \{(x,y) : \|(x,y)-(-1,0)\| \le 1\}$, $D_2 = \{(x,y) : \|(x,y)-(1,0)\| < 1\}$. They intersect at $(0,0)$. Each disk is convex and hence path-connected; their intersection is non-empty, so $D_1 \cup D_2$ is both connected and path-connected.

(b) Rational-slope lines. $A = \{(x,y) : x = 0 \text{ or } y = qx,\ q \in \mathbb{Q}\}$. Each line $y = qx$ and the y-axis are path-connected and meet at $(0,0)$. Thus their union is path-connected (hence connected).

(c) Same family minus the origin. $B = A \setminus \{(0,0)\}$. Removing $(0,0)$ disconnects the family: each branch $L_q = \{(x,y) : y = qx,\ x \neq 0\}$ and $L_0 = \{(0,y) : y \neq 0\}$ is still path-connected, but the branches are disjoint. No path can move from one branch to another without passing through $(0,0)$. Hence B is disconnected. Each component (each branch) remains path-connected.

(d) Star-shaped domain. If S is star-shaped with center x_0, then for each $x \in S$ the path

$\gamma_x(t) = (1-t)x_0 + t\,x$ ($t \in [0,1]$) lies in S, joining x_0 and x. Thus S is path-connected, hence connected.

Summary table.

Case	Description	Connected	Path-connected
(a)	Two tangent disks	Yes	Yes
(b)	Rational-slope lines (fan)	Yes	Yes
(c)	Fan without the origin	No	No (each branch yes)
(d)	Star-shaped domain	Yes	Yes

Problem 7 — Intersection of Nested Compact Connected Sets

Problem. Let $K_1 \supset K_2 \supset K_3 \supset \cdots$ be a decreasing sequence of non-empty, connected, compact subsets of a Hausdorff space X. Let

$$K = \bigcap_{n=1}^{\infty} K_n.$$

(a) Show that K is non-empty.

(b) Show that K is connected.

(c) Give an example with $X = \mathbb{R}^2$ showing that part (b) fails if the K_n are only closed instead of compact.

Theory. In any Hausdorff space, the intersection of a decreasing sequence of non-empty compact sets is non-empty (finite intersection property). Connectedness is preserved under nested intersections of connected sets. Compactness is required for non-emptiness; without it, closed sets may shrink or split into disconnected limits.

Solutions.

(a) Non-emptiness. Each K_n is compact and non-empty, $K_{n+1} \subseteq K_n$. Thus the family $\{K_n\}$ has the finite intersection property. By compactness, $\bigcap_n K_n \neq \varnothing$.

(b) Connectedness. Assume $K = A \cup B$ with A, B non-empty and disjoint, open in K. Then $A = U \cap K$, $B = V \cap K$ for some disjoint open $U, V \subset X$. For each n, $K_n = (K_n \cap U) \cup (K_n \cap V)$. Since K_n is connected, one of these intersections is empty for all n. Hence one of A, B is empty—contradiction. So K is connected.

(c) Failure without compactness. Let $X = \mathbb{R}^2$ and define

$$L_n = \{(x,y) : y = \tfrac{1}{n}x,\ x > 0\} \cup \{(x,y) : y = -\tfrac{1}{n}x,\ x > 0\}.$$

Each L_n is closed in $\mathbb{R}^2$ and connected. They form a decreasing sequence, but

$$\bigcap_n L_n = \{(x, 0) : x > 0\},$$

which is disconnected in $\mathbb{R}^2$. Thus compactness (not mere closedness) is essential.

Summary.

Part	Statement	Result	Key Idea
(a)	K non-empty	Yes	Nested compact sets $\Rightarrow$ finite intersection property
(b)	K connected	Yes	Intersection preserves connectedness
(c)	Closed instead of compact	No	May lose connectedness (counterexample in $\mathbb{R}^2$)

Problem 8 — Antipodal Points on the Circle (Borsuk–Ulam in 1D)

Problem. Show that if $f : S^1 \to \mathbb{R}$ is continuous then there exists $x \in S^1$ such that $f(-x) = f(x)$. Deduce that at any moment there exist two antipodal points on the Earth's equator with the same temperature.

Theory. This is the $n = 1$ case of the *Borsuk–Ulam theorem:* for every continuous $f : S^n \to \mathbb{R}^n$, there exists x with $f(x) = f(-x)$. For $S^1 = \{(\cos\theta, \sin\theta) : \theta \in [0, 2\pi)\}$, antipodal points correspond to θ and $\theta+\pi$. We will apply the Intermediate Value Theorem (IVT).

Solution.

Step 1. Define $g(x) = f(x) - f(-x)$. Then g is continuous and satisfies $g(-x) = -g(x)$ (odd symmetry).

Step 2. Parameterize S^1 by $x = e^{i\theta}$, $\theta \in [0, \pi]$. Then

$$g(\theta) = f(e^{i\theta}) - f(e^{i(\theta+\pi)}).$$

Note that $g(0) = -g(\pi)$.

Step 3. By the IVT, there exists $\theta_0 \in [0, \pi]$ such that $g(\theta_0) = 0$. Hence

$$f(e^{i\theta_0}) = f(e^{i(\theta_0+\pi)}),$$

so there exists $x_0 \in S^1$ with $f(x_0) = f(-x_0)$.

Step 4. Interpretation: if f measures temperature around the equator, then at any instant there are two opposite points (longitudes differing by π) having the same temperature.

Example. For $f(\theta) = \sin\theta$, we have $f(-\theta) = -f(\theta)$, so equality holds at $\theta = 0, \pi$. The theorem guarantees such antipodal equality for any continuous $f : S^1 \to \mathbb{R}$.

Problem 9 — Image of Path-connected and Compact Sets Implies Continuity

Problem. Let $f : \mathbb{R}^m \to \mathbb{R}^n$ be a function such that:

(1) the image of every path-connected subset of $\mathbb{R}^m$ is path-connected;

(2) the image of every compact subset of $\mathbb{R}^m$ is compact.

Show that f is continuous.

Theory. A continuous map between Hausdorff spaces preserves compactness and connectedness. Conversely, if a map preserves both compactness and (path-)connectedness in $\mathbb{R}^m$, then discontinuities would produce separated image pieces—contradicting path-connectedness. Hence these two preservation properties together imply continuity.

Solution.

Step 1. Assume f is not continuous at some $x_0 \in \mathbb{R}^m$. Then there exists $\varepsilon_0 > 0$ and a sequence $x_k \to x_0$ such that $\|f(x_k) - f(x_0)\| \geq \varepsilon_0$ for all k.

Step 2. Define

$$C = \{x_0\} \cup \{x_k : k \geq 1\} \cup \bigcup_{k \geq 1} [x_0, x_k],$$

where $[x_0, x_k]$ denotes the line segment joining x_0 and x_k. Then C is compact (it is bounded and closed) and path-connected.

Step 3. Consider the image $f(C)$. By assumption, $f(C)$ must be compact and path-connected. However, since $f(x_k)$ remain at least ε_0 apart from $f(x_0)$, no continuous path in $f(C)$ can join $f(x_0)$ to $f(x_k)$, contradicting path-connectedness of $f(C)$.

Step 4. Contradiction $\Rightarrow$ f is continuous at x_0. Since x_0 was arbitrary, f is continuous on $\mathbb{R}^m$.

Remarks.

- If compactness preservation alone is assumed, discontinuities may persist (e.g.

$f(x) = \text{sign}(x)$ sends compacts to compacts but breaks connectedness).

- If only connectedness is preserved, discontinuous maps can still exist (e.g. constant except at one point).
- Both properties together rule out such jumps, ensuring continuity.

Examples.

(1) Preserves compactness, not path-connectedness. Let

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then f is not continuous at 0.

Every compact $K \subset \mathbb{R}$ satisfies $f(K) \subset \{0, 1\}$, which is compact, but $f([-1, 1]) = \{0, 1\}$ is not path-connected.

(2) Preserves path-connectedness, not compactness. Let $f(x) = \tan\left(\frac{\pi}{2}x\right)$.

Between vertical asymptotes f is continuous and monotone, so the image of any connected interval is an interval (path-connected).

However, $f([0, 1])$ is unbounded, hence not compact.

Thus f is discontinuous and fails compactness preservation.

(3) Both properties $\Rightarrow$ continuity. Any discontinuous function must create a "jump" separating image points, breaking path-connectedness or compactness of some compact, path-connected set.

Therefore a map preserving both properties must be continuous.

Problem 10 — Differentiability and Continuity via Partial Derivatives

Problem. Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $(x_0, y_0) \in \mathbb{R}^2$.

(a) Suppose $\partial_1 f$ exists and is continuous in an open ball around (x_0, y_0), and $\partial_2 f$ exists at (x_0, y_0). Show that f is differentiable at (x_0, y_0).

(b) Suppose instead $\partial_1 f$ exists and is bounded in an open ball around (x_0, y_0), and that for fixed x, the map $y \mapsto f(x, y)$ is continuous. Show that f is continuous at (x_0, y_0).

Theory. A function $f : \mathbb{R}^2 \to \mathbb{R}$ is differentiable at (x_0, y_0) if there exists a linear map $L(h, k) = ah + bk$ with

$$\lim_{(h,k)\to(0,0)} \frac{f(x_0 + h, y_0 + k) - f(x_0, y_0) - ah - bk}{\sqrt{h^2 + k^2}} = 0.$$

Continuous partials near a point guarantee differentiability there. Bounded partials plus separate continuity imply overall continuity.

Solution.

(a) Differentiability. Write

$$f(x_0+h, y_0+k) - f(x_0, y_0) = [f(x_0+h, y_0+k) - f(x_0, y_0+k)] + [f(x_0, y_0+k) - f(x_0, y_0)].$$

By the Mean Value Theorem in x, $f(x_0 + h, y_0 + k) - f(x_0, y_0 + k) = h\,\partial_1 f(\xi, y_0 + k)$ for some $\xi \in (x_0, x_0 + h)$. Thus

$$f(x_0+h, y_0+k) - f(x_0, y_0) = h\,\partial_1 f(x_0, y_0) + k\,\partial_2 f(x_0, y_0) + h[\partial_1 f(\xi, y_0+k) - \partial_1 f(x_0, y_0)] + o(k).$$

Continuity of $\partial_1 f$ implies the last term is $o(\sqrt{h^2 + k^2})$, so f is differentiable at (x_0, y_0) with

$$Df(x_0, y_0) = (\partial_1 f(x_0, y_0),\ \partial_2 f(x_0, y_0)).$$

(b) Continuity. For (x, y) near (x_0, y_0),

$$f(x, y) - f(x_0, y_0) = [f(x, y) - f(x_0, y)] + [f(x_0, y) - f(x_0, y_0)].$$

By the Mean Value Theorem, $f(x, y) - f(x_0, y) = (x - x_0)\partial_1 f(\xi, y)$ for some ξ between x_0 and x. If $|\partial_1 f| \le M$, then $|f(x, y) - f(x_0, y)| \le M|x - x_0|$. Hence

$$|f(x, y) - f(x_0, y_0)| \le M|x - x_0| + |f(x_0, y) - f(x_0, y_0)|.$$

The second term $\to 0$ as $y \to y_0$, and the first $\to 0$ as $x \to x_0$, so f is continuous at (x_0, y_0).

Examples.

- $f(x, y) = x^2 + y^2$ — all partials continuous $\Rightarrow$ differentiable everywhere.
- $f(x, y) = x|y|$ — $\partial_1 f = y|y|/|y| = |y|$ is bounded, and $y \mapsto f(x, y)$ is continuous for each fixed $x \Rightarrow f$ is continuous at $(0, 0)$.

Problem 11 — Differentiability of $f(x) = x/\|x\|$

Problem. Consider the map $f : \mathbb{R}^n \to \mathbb{R}^n$ given by

$$f(x) = \frac{x}{\|x\|} \text{ for } x \neq 0, \qquad f(0) = 0.$$

(a) Using the definition of derivative, show that $x \mapsto \|x\|^2$ is differentiable and compute its derivative. *(Avoid using partial derivatives.)*

(b) Using part (a) and questions 5(b)–5(c), show that f is differentiable except at 0 and that

$$Df|_x(h) = \frac{h}{\|x\|} - \frac{x(x\cdot h)}{\|x\|^3}.$$

(Avoid using partial derivatives.)

(c) Verify that $Df|_x(h)$ is orthogonal to x and explain geometrically why.

Theory. Differentiability in $\mathbb{R}^n$ means

$$f(x+h) - f(x) = Df|_x(h) + o(\|h\|).$$

For $g(x) = \langle x, x\rangle = \|x\|^2$ we use the product rule of the inner product. For $f(x) = x/\|x\|$ we use the scalar–vector product and the chain rule.

Solution.

(a) Differentiability of $\|x\|^2$.

$$g(x+h) - g(x) = \|x+h\|^2 - \|x\|^2 = 2\langle x, h\rangle + \|h\|^2.$$

Hence

$$Dg|_x(h) = 2\langle x, h\rangle.$$

Therefore g is differentiable everywhere, with gradient $\nabla g(x) = 2x$.

(b) Differentiability of $f(x) = x/\|x\|$ for $x \neq 0$. Let $r(x) = \|x\|$, so that $f(x) = x\, r(x)^{-1}$. From part (a),

$$Dr|_x(h) = \frac{\langle x, h\rangle}{\|x\|}.$$

By the product rule,

$$Df|_x(h) = h\,\frac{1}{\|x\|} + x\, D(r^{-1})|_x(h).$$

Since $D(r^{-1})|_x(h) = -\|x\|^{-2} Dr|_x(h) = -\frac{\langle x,h\rangle}{\|x\|^3}$, we get

$$Df|_x(h) = \frac{h}{\|x\|} - \frac{x(x\cdot h)}{\|x\|^3}.$$

At $x = 0$, f is not continuous (directional limits depend on direction), hence not differentiable there.

(c) Orthogonality. Compute

$$x\cdot Df|_x(h) = \frac{x\cdot h}{\|x\|} - \frac{(x\cdot x)(x\cdot h)}{\|x\|^3} = 0.$$

Thus $Df|_x(h) \perp x$.

Geometric meaning. $f(x)$ projects x onto the unit sphere. The derivative $Df|_x$ gives the orthogonal projection of the displacement h onto the tangent hyperplane to the sphere at $f(x)$. Hence its image lies entirely in that tangent plane, perpendicular to x.

Problem 12 — Differentiability and Continuous Square Roots on M_n

Problem. Define $f : M_n \to M_n$ by $f(A) = A^2$.

(a) Show that f is continuously differentiable on the whole of M_n.

(b) Deduce that there is a continuous square-root function on some neighbourhood of the identity Id; that is, show that there exists an open ball $B_\varepsilon(\mathrm{Id})$ and a continuous function $g : B_\varepsilon(\mathrm{Id}) \to M_n$ such that $g(A)^2 = A$ for all $A \in B_\varepsilon(\mathrm{Id})$.

(c) Is it possible to define a continuous square-root function on the whole of M_n?

Theory. Matrix multiplication $(A, B) \mapsto AB$ is bilinear and continuous on $M_n \times M_n$. Hence polynomial maps in A are continuously differentiable. The derivative of $A \mapsto A^2$ at A is given by

$$Df|_A(H) = AH + HA.$$

The Inverse Function Theorem in Banach spaces guarantees a local inverse of f near points where $Df|_A$ is invertible.

Solution.

(a) Differentiability. For $A, H \in M_n$,

$$f(A + H) - f(A) = (A + H)^2 - A^2 = AH + HA + H^2.$$

Hence

$$\frac{\|f(A + H) - f(A) - (AH + HA)\|}{\|H\|} = \frac{\|H^2\|}{\|H\|} \to 0.$$

So $Df|_A(H) = AH + HA$. Since

$$\|Df|_{A_1} - Df|_{A_2}\| \leq 2\|A_1 - A_2\|,$$

the derivative map is continuous, and $f \in C^1(M_n, M_n)$.

(b) Continuous square root near Id. At Id we have $f(\mathrm{Id}) = \mathrm{Id}$ and $Df|_{\mathrm{Id}}(H) = 2H$, which is invertible. By the Banach-space Inverse Function Theorem, there exists $\varepsilon > 0$ and a continuous map

$$g : B_\varepsilon(\mathrm{Id}) \to M_n$$

such that $g(A)^2 = A$ for all $A \in B_\varepsilon(\mathrm{Id})$. Thus a continuous matrix square-root exists locally near Id.

(c) Nonexistence of a global continuous square root. No continuous $g : M_n \to M_n$ with $g(A)^2 = A$ can exist globally. Indeed, $\det(g(A))^2 = \det(A)$. If A crosses from $\det(A) > 0$ to $\det(A) < 0$, $\det(g(A))$ must change sign, forcing a discontinuity. Hence no continuous global square-root function can exist on M_n.

Differentiable vs Continuously Differentiable Functions

Definition (Differentiable). A map $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at x_0 if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h\to 0} \frac{\|f(x_0+h) - f(x_0) - L(h)\|}{\|h\|} = 0.$$

Then $L = Df(x_0)$ is the (Fréchet) derivative at x_0.

Definition (Continuously differentiable). f is continuously differentiable (C^1) on a domain if it is differentiable at every point and the derivative map $x \mapsto Df(x)$ is continuous.

Distinction. Differentiability only ensures that a linear approximation exists pointwise. Continuous differentiability requires that this linear map depends continuously on the point.

Example 1 (Smooth case). $f(x) = x^2$ is differentiable everywhere with $f'(x) = 2x$, continuous. Hence $f \in C^1(\mathbb{R})$.

Example 2 (Differentiable but not C^1).

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then

$$f'(x) = \begin{cases} 2x \sin\frac{1}{x} - \cos\frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

f is differentiable at 0 since $\lim_{x\to 0} \frac{f(x)-f(0)}{x} = 0$, but f' oscillates between -1 and 1, not continuous at 0. Thus f is differentiable but not continuously differentiable.

Example 3 (Derivative unbounded).

$$f(x) = x^{1/3}, \quad f'(x) = \frac{1}{3}x^{-2/3}.$$

Derivative exists for $x \neq 0$ but is unbounded near 0, hence not continuous there.

Summary table.

Property	Meaning	Implication
Differentiable at x_0	Linear approximation exists	Local condition
Differentiable everywhere	Derivative exists at all points	May be discontinuous
Continuously differentiable (C^1)	Derivative continuous	Smooth, stable variation
$C^1 \Rightarrow$ differentiable	Always true	
Differentiable $\not\Rightarrow C^1$	Counterexamples exist	

Geometric intuition. For a differentiable but not C^1 function, the tangent line exists at each point, but its slope changes abruptly. Continuous differentiability ensures the tangent depends smoothly on the point.

Problem 13 — Differentiability of the Determinant Function

Problem. Consider $\det : GL_n(\mathbb{R}) \to \mathbb{R}$, where

$$GL_n(\mathbb{R}) = \{X \in M_n : \ X \text{ invertible}\}.$$

(a) Show that $GL_n(\mathbb{R})$ is an open subset of M_n.

(b) Show that det is differentiable on all of $GL_n(\mathbb{R})$ and that

$$D(\det)|_A(H) = \det(A)\,\mathrm{tr}(A^{-1}H).$$

(c) Show that det is twice differentiable at I and find $D^2(\det)|_I$ as a bilinear map.

Theory. The determinant is a polynomial in the entries of A, hence smooth. For $A \in GL_n$, we may compute its derivative intrinsically using the Jacobi identity:

$$\frac{d}{dt}\det(A+tH) = \det(A+tH)\,\mathrm{tr}((A+tH)^{-1}H).$$

Matrix multiplication and inversion are smooth on GL_n.

Solution.

(a) Openness. The determinant map $\det : M_n \to \mathbb{R}$ is continuous, and $GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$. Since $\mathbb{R} \setminus \{0\}$ is open, its preimage is open. Thus $GL_n(\mathbb{R})$ is open in M_n.

(b) First derivative. Let $A \in GL_n$ and $H \in M_n$. Define $\phi(t) = \det(A + tH)$. Then

$$\phi'(t) = \det(A + tH)\,\mathrm{tr}((A + tH)^{-1}H),$$

so

$$D(\det)|_A(H) = \phi'(0) = \det(A)\,\mathrm{tr}(A^{-1}H).$$

Hence det is differentiable on GL_n.

(c) Second derivative at I. At $A = I$, $\det(I) = 1$ and $D(\det)|_I(H) = \mathrm{tr}(H)$. For A near I,

$$D(\det)|_A(H) = \det(A)\,\mathrm{tr}(A^{-1}H).$$

Differentiate with respect to A in direction K:

$$D^2(\det)|_I(K, H) = D(\det)|_I(K)\,\mathrm{tr}(H) + \det(I)\,\mathrm{tr}(D(A^{-1})|_I(K)\,H).$$

Since $D(A^{-1})|_I(K) = -K$, we obtain

$$D^2(\det)|_I(K, H) = \mathrm{tr}(K)\mathrm{tr}(H) - \mathrm{tr}(KH).$$

This defines a symmetric bilinear form on M_n.

Geometric interpretation. $D(\det)|_A(H)$ measures infinitesimal relative volume change in direction H. At I, the second derivative

$$D^2(\det)|_I(K, H) = \mathrm{tr}(K)\mathrm{tr}(H) - \mathrm{tr}(KH)$$

describes curvature of the volume function at the identity. The determinant is actually C^∞ on $GL_n(\mathbb{R})$.

Problem 14 — Convergence of Series of Functions

Problem. Let $f_n : I \to \mathbb{R}$ be given by

$$\begin{aligned}
&\text{(a)}\quad f_n(x) = xe^{-nx}, \quad I = [0, +\infty);\\
&\text{(b)}\quad f_n(x) = \frac{n^2 x}{1 + n^4 x^2}, \quad I = [0, 1];\\
&\text{(c)}\quad f_n(x) = (-1)^n \frac{x^n}{n}, \quad I = [0, 1].
\end{aligned}$$

Determine for each case whether $\sum_{n=1}^{\infty} f_n(x)$ converges or diverges on I, and if it converges, whether the convergence is pointwise, uniform, pointwise absolute, or uniform absolute.

Theory. For a series of functions $\sum f_n$,

- it converges *pointwise* if for each x the numeric series $\sum f_n(x)$ converges;

- it converges *uniformly* if $\sup_{x\in I} |\sum_{k>n} f_k(x)| \to 0$;
- it is *absolutely convergent* if $\sum |f_n(x)|$ converges;
- *uniformly absolute* if $\sum \|f_n\|_\infty < \infty$.

(a) $f_n(x) = xe^{-nx}$ on $[0, +\infty)$.

For fixed $x > 0$,

$$\sum_{n=1}^{\infty} f_n(x) = x \sum_{n=1}^{\infty} e^{-nx} = x \frac{e^{-x}}{1 - e^{-x}},$$

which converges for all $x \geq 0$. At $x = 0$, $f_n(0) = 0$. Hence the series converges pointwise and absolutely on $[0, +\infty)$.

Since $\|f_n\|_\infty = \frac{1}{ne}$, the convergence is not uniform on the whole interval, but it is uniform on each compact $[0, M]$.

Conclusion: Pointwise and absolute convergence, uniform on compacts but not on $[0, +\infty)$.

(b) $f_n(x) = \dfrac{n^2 x}{1 + n^4 x^2}$ on $[0, 1]$.

For each fixed $x > 0$, $f_n(x) \sim \frac{1}{n^2 x} \to 0$, so the series $\sum f_n(x)$ converges pointwise and absolutely. At $x = 0$, $f_n(0) = 0$.

However, as $x \to 0$, the bound $\frac{1}{n^2 x}$ blows up, so convergence is not uniform on $[0, 1]$. It is uniform on every subinterval $[\delta, 1]$, $\delta > 0$.

Conclusion: Pointwise and absolute convergence, uniform on $[\delta, 1]$ but not on $[0, 1]$.

(c) $f_n(x) = (-1)^n \dfrac{x^n}{n}$ on $[0, 1]$.

For $x \in [0, 1)$,

$$\sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} (-1)^n \frac{x^n}{n} = \ln(1 + x) - x,$$

which converges for all $x \in [0, 1)$ and also at $x = 1$ (to $-\ln 2$). By the Leibniz criterion, the convergence is uniform on $[0, 1]$.

The absolute series $\sum |f_n(x)| = \sum x^n / n$ diverges at $x = 1$ but converges for $x < 1$. Thus the convergence is not uniformly absolute.

Conclusion: Uniformly convergent on $[0, 1]$, pointwise absolutely convergent for $x < 1$ but not uniformly absolute.

Summary.

Case	Pointwise	Uniform	Absolute	Uniform absolute
(a) xe^{-nx}	✓	on compacts	✓	$\times$
(b) $\dfrac{n^2x}{1+n^4x^2}$	✓	$[\delta, 1]$	✓	$\times$
(c) $(-1)^n\dfrac{x^n}{n}$	✓	✓	for $x < 1$	$\times$

Sequences vs Series of Functions — Definitions

Sequences of functions. Let X be a set and Y a metric space (e.g. $\mathbb{R}$). A sequence of functions is a list $(f_n)_{n\geq 1}$ with $f_n : X \to Y$.

- *Pointwise convergence:* $f_n \to f$ on X if $\lim_{n\to\infty} f_n(x) = f(x)$ for every $x \in X$.
- *Uniform convergence:* $f_n \to f$ uniformly on X if

$$\sup_{x\in X} |f_n(x) - f(x)| \xrightarrow[n\to\infty]{} 0.$$

- *Uniform Cauchy:* (f_n) is uniformly Cauchy if $\sup_{x\in X} |f_m(x) - f_n(x)| \to 0$ as $m, n \to \infty$. If Y is complete, this is equivalent to uniform convergence.

Series of functions. Given (f_n), define partial sums $S_N(x) = \sum_{n=1}^{N} f_n(x)$. The series $\sum_{n=1}^{\infty} f_n$ *converges pointwise* to S if $S_N(x) \to S(x)$ for all $x \in X$. It *converges uniformly* if $\sup_{x\in X} |S_N(x) - S(x)| \to 0$.

- *Pointwise absolute convergence:* $\sum |f_n(x)|$ converges for each x.
- *Uniform absolute convergence:* $\sum \|f_n\|_\infty < \infty$. (By the Weierstrass M-test, this implies uniform convergence of $\sum f_n$.)

Examples.

- *Sequence only.* $f_n(x) = x^n$ on $[0, 1]$: $f_n \to \mathbf{1}_{\{1\}}$ pointwise, not uniformly.
- *Series.* $f_n(x) = (-1)^n x^n/n$ on $[0, 1]$: $\sum f_n$ converges uniformly (alternating test), but not absolutely at $x = 1$ since $\sum |f_n(1)| = \sum 1/n$ diverges.

Remark. A series of functions $\sum f_n$ converges iff its partial sums (S_N) (a *sequence of functions*) converge.

Problem 15 — The series $\sum_{n=1}^{\infty}(x - n)^{-2}$ on $X = \mathbb{R} \setminus \mathbb{N}$

Problem. Consider the series $\sum_{n=1}^{\infty}(x - n)^{-2}$ for $x \in X := \mathbb{R} \setminus \mathbb{N}$.

(a) Show that the series converges pointwise on X.

(b) Does the series converge uniformly on X?

(c) Show that the sum $f(x) = \sum_{n=1}^{\infty}(x-n)^{-2}$ is continuous on X.

(a) Pointwise convergence. Fix $x \in X$. For $n > |x|+1$, $|x-n| \geq n - |x| \geq \frac{n}{2}$, hence

$$0 \leq \frac{1}{(x-n)^2} \leq \frac{4}{n^2}.$$

Since $\sum 4/n^2$ converges, by comparison $\sum (x-n)^{-2}$ converges (absolutely) for the fixed x. Thus the series converges pointwise on X.

(b) Not uniformly convergent on X. Uniform convergence would imply

$$\sup_{x\in X} \sum_{n=N}^{M} \frac{1}{(x-n)^2} \xrightarrow[N,M\to\infty]{} 0.$$

But for any N choose an integer $k \geq N$ and take $x = k+\delta$ with $0 < \delta < 1$. Then the tail contains the term $1/(x-k)^2 = 1/\delta^2$, which can be made arbitrarily large as $\delta \downarrow 0$. Hence the supremum of the tail is infinite for every N, so the Cauchy criterion fails. Therefore the series is *not* uniformly convergent on X.

(c) Continuity of the sum. Fix $x_0 \in X$ and set $\delta = \frac{1}{2}\operatorname{dist}(x_0, \mathbb{N}) > 0$. On the interval $I = (x_0-\delta, x_0+\delta) \subset X$, the finitely many functions $(x-n)^{-2}$ with n near I are continuous, and for all sufficiently large n and all $x \in I$,

$$\frac{1}{(x-n)^2} \leq \frac{4}{n^2}.$$

By the Weierstrass M-test, the tail $\sum_{n>N}(x-n)^{-2}$ converges uniformly on I. Hence the whole series converges uniformly on I to a continuous function. Since x_0 was arbitrary, the sum f is continuous on X.

Conclusion. The series converges pointwise and absolutely on X, is not uniformly convergent on X, and its sum belongs to $C(X)$ (local uniform convergence).

Convergence Criteria for Sequences and Series of Functions

A) Sequences of functions $(f_n : X \to \mathbb{R})$**.**

- *Pointwise convergence:* $f_n \to f$ if $\forall x \in X,\ f_n(x) \to f(x)$.
- *Uniform convergence:* $f_n \to f$ uniformly if

$$\sup_{x\in X} |f_n(x) - f(x)| \xrightarrow[n\to\infty]{} 0.$$

- *Uniform Cauchy criterion:* f_n converges uniformly $\iff$ $\sup_{x\in X}|f_m(x)-f_n(x)| \xrightarrow[m,n\to\infty]{} 0$.
- *Dini (compact X):* If X compact, $f_n \in C(X)$, $f_n \downarrow f$ or $f_n \uparrow f$ pointwise and $f \in C(X)$, then $f_n \to f$ uniformly.
- *Uniform limit theorem:* If $f_n \in C(X)$ and $f_n \to f$ uniformly, then $f \in C(X)$.

Examples.

- $f_n(x) = x^n$ on $[0,1]$: pointwise $\to \mathbf{1}_{\{1\}}$, not uniform.
- Geometric partial sums $S_n(x) = \sum_{k=0}^{n} x^k$ on $[0,a]$ with $0 < a < 1$: (S_n) is uniformly Cauchy $\Rightarrow$ uniform convergence.
- $f_n(x) = x^n$ on $[0,a]$ with $0 < a < 1$: monotone $\downarrow$ to $0 \in C([0,a]) \Rightarrow$ uniform by Dini.

B) Series of functions $\sum_{n=1}^{\infty} f_n$.

Let $S_N = \sum_{n=1}^{N} f_n$.

- *Pointwise convergence:* $S_N(x) \to S(x)$ for each $x \in X$.
- *Uniform convergence (Cauchy form):* $\sum f_n$ converges uniformly $\iff$ $\sup_{x\in X}\left|\sum_{k=m}^{n} f_k(x)\right| \xrightarrow[m,n\to\infty]{} 0$.
- *Weierstrass M-test (uniform absolute):* If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly (hence absolutely).
- *Leibniz alternating test (uniform):* If $f_n = (-1)^n a_n$ with $a_n \ge 0$, $a_{n+1}(x) \le a_n(x)$ for all x and $a_n \to 0$ uniformly on X, then $\sum f_n$ converges uniformly on X.
- *Dirichlet test (uniform):* If partial sums $A_N(x) = \sum_{n=1}^{N} a_n(x)$ are uniformly bounded on X, and $b_n(x)$ is monotone in n for each x with $b_n \to 0$ uniformly, then $\sum a_n(x)b_n(x)$ converges uniformly.
- *Abel test (uniform):* If $\sum a_n(x)$ converges uniformly on X and $b_n(x)$ is uniformly bounded and monotone in n for each x, then $\sum a_n(x)b_n(x)$ converges uniformly.
- *Preservation:* If $f_n \in C(X)$ and $\sum f_n$ converges uniformly, then the sum $S \in C(X)$ (and $\int \sum f_n = \sum \int f_n$).

Examples.

- (*M-test*) $f_n(x) = \dfrac{x^n}{n^2}$ on $[0,1]$: $|f_n| \le 1/n^2$, $\sum 1/n^2 < \infty \Rightarrow$ uniform absolute convergence.
- (*Leibniz*) $f_n(x) = (-1)^n \dfrac{x^n}{n}$ on $[0,1]$: $a_n = x^n/n \downarrow$ and $a_n \to 0$ uniformly $\Rightarrow$ uniform convergence (not uniformly absolute).

- (*Dirichlet*) $\sum_{n=1}^{\infty} \dfrac{\sin(nx)}{n}$ on $[\delta, 2\pi - \delta]$: partial sums of $\sin(nx)$ uniformly bounded; $1/n \downarrow 0$ uniformly $\Rightarrow$ uniform convergence on those compact subintervals.
- (*Pointwise but not uniform*) $\sum_{n=1}^{\infty} \dfrac{n^2 x}{1+n^4x^2}$ on $[0,1]$: pointwise (absolute) convergence, not uniform on the whole interval.
- (*Locally uniform*) $\sum_{n=1}^{\infty}(x-n)^{-2}$ on $X = \mathbb{R} \setminus \mathbb{N}$: not uniform on X, but uniform on neighborhoods avoiding integers $\Rightarrow$ sum is continuous.

What Uniform Convergence Buys You (and Key Tools)

Uniform Limit Theorem. If X is a metric space, $f_n \in C(X)$, and $f_n \to f$ uniformly on X, then $f \in C(X)$.

Integration commutes with uniform limit (Riemann). If $f_n : [a,b] \to \mathbb{R}$ are Riemann integrable and $f_n \to f$ uniformly on $[a,b]$, then

$$\int_a^b f_n \, dx \to \int_a^b f \, dx.$$

Termwise integration of series. If $\sum f_n$ converges uniformly on $[a,b]$ and each f_n is Riemann integrable, then

$$\int_a^b \left(\sum_{n=1}^{\infty} f_n \right) = \sum_{n=1}^{\infty} \int_a^b f_n.$$

Termwise differentiation (sufficient condition). Let $I = [a,b]$. If $f_n \in C^1(I)$, $f_n' \to g$ uniformly on I, and $(f_n(x_0))_n$ converges for some $x_0 \in I$, then $f_n \to f$ uniformly on I, $f \in C^1(I)$, and $f' = g$.

Power series. For $\sum_{n=0}^{\infty} a_n (x-x_0)^n$ with radius $R > 0$: uniform convergence on each closed $[x_0 - r, x_0 + r] \subset (x_0 - R, x_0 + R)$; termwise integration/differentiation valid on $(x_0 - R, x_0 + R)$; the derivative series has radius R.

Dini's theorem (compact X). If X is compact, $f_n \in C(X)$, and $f_n \downarrow f \in C(X)$ (or $f_n \uparrow f$), then $f_n \to f$ uniformly.

Arzelà–Ascoli. If X is compact and $\mathcal{F} \subset C(X)$ is equicontinuous and pointwise bounded, then every sequence in $\mathcal{F}$ has a uniformly convergent subsequence with continuous limit.

Uniform Dirichlet test. If $A_N(x) = \sum_{n=1}^{N} a_n(x)$ are uniformly bounded on X, and $b_n(x)$ is monotone in n for each x with $b_n \to 0$ uniformly, then $\sum a_n(x) b_n(x)$ converges uniformly.

Uniform Abel test. If $\sum a_n(x)$ converges uniformly on X and $b_n(x)$ is uniformly bounded and monotone in n for each x, then $\sum a_n(x) b_n(x)$ converges uniformly.

Mini-examples.

- (Uniform limit) $f_n(x) = x^n$ on $[0, a]$, $0 < a < 1$: $f_n \downarrow 0$ and $f_n \to 0$ uniformly (Dini).
- (M-test) $\sum \frac{x^n}{n^2}$ on $[0, 1]$: $|f_n| \le 1/n^2$, so uniform absolute convergence; integrals may be swapped with sum.
- (Alternating uniform) $\sum (-1)^n \frac{x^n}{n}$ on $[0, 1]$: uniform (Leibniz), but not uniformly absolute.
- (Dirichlet) $\sum \frac{\sin(nx)}{n}$: uniform on $[\delta, 2\pi - \delta]$ for any $\delta > 0$.
- (Arzelà–Ascoli) $f_n(x) = \sin(nx)/n$ on $[0, 2\pi]$: equicontinuous and bounded $\Rightarrow$ a uniformly convergent subsequence.

Gotchas. Pointwise convergence does not preserve continuity or integrals; uniform convergence does.

Uniform absolute $\Rightarrow$ uniform, but not conversely.

Local uniform convergence suffices for continuity on open domains.

Decision Tree for Convergence of Sequences and Series of Functions

Series of functions $\sum f_n$ on X.

1) *Uniform bound by a summable scalar?*
   If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty \Rightarrow$ **Weierstrass M-test**: uniform absolute convergence.
2) *Alternating in n? $f_n = (-1)^n a_n$.*
   If $a_n(x) \ge 0$, nonincreasing in n for each x, and $a_n \to 0$ uniformly on $X \Rightarrow$ **Leibniz (uniform)**: uniform convergence (typically not uniformly absolute).
3) *Product form $a_n(x)b_n(x)$?*
   Dirichlet (uniform): if $A_N(x) = \sum_{n=1}^{N} a_n(x)$ are uniformly bounded on X and $b_n(x)$ is monotone in n with $b_n \to 0$ uniformly, then $\sum a_n b_n$ converges uniformly.
   Abel (uniform): if $\sum a_n(x)$ converges uniformly on X and $b_n(x)$ is uniformly bounded and monotone in n, then $\sum a_n b_n$ converges uniformly.
4) *Power series / radius present?*
   Determine radius R via root/ratio pointwise. Convergence is uniform on every closed subinterval strictly inside the radius; termwise differentiation/integration valid in $(x_0 - R, x_0 + R)$.
5) *Spikes or singularities at isolated points?*
   Expect global uniform *fails.* Prove **local uniform** convergence on neighborhoods avoiding singularities to deduce continuity on the punctured domain.

6) *No pattern? Use uniform Cauchy on compacta.*
   Show tail $\sup_{x\in K} |\sum_{n>N} f_n(x)| \to 0$ for each compact $K \subset X$.
7) *Absolute vs conditional.*
   If $\sum \|f_n\|_\infty < \infty \Rightarrow$ uniform absolute. Alternating/Dirichlet/Abel often give uniform but not absolute.

Sequences of functions (f_n).

1) **Dini (compact X):** If $f_n \in C(X)$ and $f_n \downarrow f \in C(X)$ (or $\uparrow$), then $f_n \to f$ uniformly.
2) **Arzelà–Ascoli:** On compact X, equicontinuous and pointwise bounded families have uniformly convergent subsequences.
3) **Uniform Cauchy:** $\sup_{x\in X} |f_m(x) - f_n(x)| \to 0 \Rightarrow$ uniform convergence.
4) **Power-series tails:** Use M-test on tails to get uniform convergence on compacta.

Micro-examples.

- (M-test) $f_n(x) = \dfrac{x^n}{n^2}$ on $[0,1]$: $|f_n| \le 1/n^2$, so uniform absolute convergence.
- (Leibniz) $f_n(x) = (-1)^n \dfrac{x^n}{n}$ on $[0,1]$: $a_n = x^n/n$ decreases in n, $a_n \to 0$ uniformly $\Rightarrow$ uniform, not uniformly absolute.
- (Dirichlet) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(nx)}{n}$: uniform on $[\delta, 2\pi - \delta]$ for any $\delta > 0$; not uniform on all of $[0, 2\pi]$.
- (Local uniform) $\displaystyle\sum_{n=1}^{\infty} (x-n)^{-2}$ on $\mathbb{R} \setminus \mathbb{N}$: not uniform globally; locally uniform $\Rightarrow$ continuous sum on the punctured domain.
- (Dini) $f_n(x) = x^n$ on $[0,a]$, $0 < a < 1$: $f_n \downarrow 0 \in C \Rightarrow$ uniform.

What uniform convergence buys you. Uniform limit of continuous functions is continuous; integrals pass to the limit. For series: uniform convergence $\Rightarrow$ termwise integration; with extra hypotheses on derivatives, termwise differentiation on compact intervals.

Problem 16 — Implicit Function Theorem via a block map

Setup. Let $U \subset \mathbb{R}^{n+m}$ be open and $f : U \to \mathbb{R}^m$ be C^1. Write $(x,y) \in \mathbb{R}^n \times \mathbb{R}^m$ and assume $(x_0, y_0) \in U$ with

$$f(x_0, y_0) = 0 \qquad \text{and} \qquad J_y f\Big|_{(x_0,y_0)} := \det\left[\frac{\partial f^i}{\partial y^j}\right]_{i,j=1}^{m} \neq 0.$$

(a) Local invertibility of F. Define $F(x, y) = (x, f(x, y))$. Its Jacobian at (x_0, y_0) is

$$DF\Big|_{(x_0,y_0)} = \begin{pmatrix} I_n & 0 \\ D_x f\Big|_{(x_0,y_0)} & D_y f\Big|_{(x_0,y_0)} \end{pmatrix}.$$

This block–triangular matrix has $\det DF = \det(I_n)\,\det(D_y f) \neq 0$, so by the inverse function theorem F is a local C^1 diffeomorphism near (x_0, y_0).

(b) A property of the inverse G. Let $G = (G^1, \ldots, G^{n+m})$ be the local inverse of F on a neighborhood $V \ni (x_0, 0)$. For any $(X, 0) \in V$,

$$F\Big(G(X,0)\Big) = (X,0) \quad \Longrightarrow \quad \begin{cases} (G^1, \ldots, G^n)(X,0) = X, \\ f\Big(X,\ G^{n+1}(X,0), \ldots, G^{n+m}(X,0)\Big) = 0. \end{cases}$$

(c) Existence and uniqueness of the implicit function. Define, for X near x_0,

$$g(X) := \Big(G^{n+1}(X,0), \ldots, G^{n+m}(X,0)\Big).$$

Then $f(X, g(X)) = 0$ and $g(x_0) = y_0$. If $f(X, y) = 0$ with X near x_0, y near y_0, then $F(X, y) = (X, 0)$, so $(X, y) = G(X, 0)$ and hence $y = g(X)$ (uniqueness).

(d) Differentiability and the formula for Dg. Since G is C^1, so is g. Differentiating $f(x, g(x)) \equiv 0$ gives

$$D_x f(x, g(x)) \ + \ D_y f(x, g(x))\, Dg(x) \ = \ 0,$$

and because $D_y f$ stays invertible nearby,

$$\boxed{Dg(x) \ = \ -\Big[D_y f(x, g(x))\Big]^{-1} D_x f(x, g(x))}.$$

(e) Application: the folium $x^3 + y^3 - 3xy = 0$. Let $f(x, y) = x^3 + y^3 - 3xy$. Then

$$\frac{\partial f}{\partial y} = 3y^2 - 3x, \qquad \frac{\partial f}{\partial x} = 3x^2 - 3y.$$

On the level set $C = \{f = 0\}$ the IFT yields a local graph $y = g(x)$ at points with $\frac{\partial f}{\partial y} \neq 0$. Solving on C the system $\{f = 0,\ \partial f/\partial y = 0\}$ gives $x = y^2$ and $y^6 - 2y^3 = 0$, so $(x, y) = (0, 0)$ or $(2^{2/3}, 2^{1/3})$. Hence C is locally a graph $y = g(x)$ away from small neighborhoods of these two exceptional points.

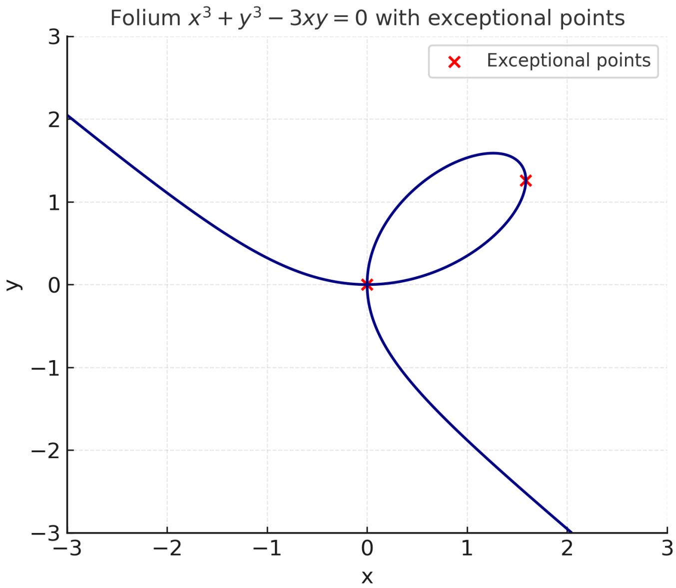

Figure 1: The folium $x^3 + y^3 - 3xy = 0$ with the two exceptional points marked: $(0, 0)$ and $(2^{2/3}, 2^{1/3})$. At these points $\partial f/\partial y = 0$ on the curve, so the implicit function theorem does not yield a local graph $y = g(x)$.

Here is a visualization of the *folium curve*

$$x^3 + y^3 - 3xy = 0,$$

showing the **exceptional points** $(0, 0)$ and $(2^{2/3}, 2^{1/3})$ where IFT fails for $y = g(x)$. Indeed,

$$\frac{\partial f}{\partial y} = 3y^2 - 3x = 0 \iff x = y^2, \qquad f(x, y) = 0 \iff y^6 - 2y^3 = 0 \iff y = 0 \text{ or } y^3 = 2.$$

Everywhere else on the curve, $\partial f/\partial y \neq 0$ and the set is locally a graph $y = g(x)$.

Implicit Function Theorem — statement, intuition, examples

IFT (finite-dimensional C^1). Let $U \subset \mathbb{R}^{n+m}$ be open and $f : U \to \mathbb{R}^m$ be C^1. Write $(x, y) \in \mathbb{R}^n \times \mathbb{R}^m$. Suppose $(x_0, y_0) \in U$ with

$$f(x_0, y_0) = 0, \qquad \det D_y f(x_0, y_0) \neq 0.$$

Then there exist neighborhoods $W' \ni x_0$, $W'' \ni y_0$ and a unique map $g : W' \to W''$, of class C^1, such that $g(x_0) = y_0$ and $f(x, g(x)) = 0$ on W'. Moreover

$$Dg(x) = -\left[D_y f(x, g(x))\right]^{-1} D_x f(x, g(x)).$$

If $f \in C^k$ (resp. real-analytic) then $g \in C^k$ (resp. analytic).

Geometric picture. Near (x_0, y_0), the zero set $f^{-1}(0)$ is a smooth n-dimensional surface; in suitable coordinates it is the graph $y = g(x)$.

Proof sketch. Apply the inverse function theorem to $F(x, y) = (x, f(x, y))$:

$$DF\Big|_{(x_0,y_0)} = \begin{pmatrix} I_n & 0 \\ D_x f & D_y f \end{pmatrix}_{(x_0,y_0)}$$

is invertible by hypothesis (block-triangular). Then F has a local inverse G and $g(x) := (G^{n+1}, \ldots, G^{n+m})(x, 0)$.

Examples.

1) Circle. $f(x, y) = x^2 + y^2 - 1$. At $(0, 1)$, $\partial f/\partial y = 2 \neq 0$, so $y = g(x)$ exists locally (upper semicircle). At $(1, 0)$ we cannot solve for y, but can solve for $x = h(y)$ since $\partial f/\partial x = 2 \neq 0$.

2) Parabola: solve for the right variable. $f(x, y) = y^2 - x$ at $(0, 0)$ has $\partial f/\partial y = 0$ (no $y = g(x)$), but $\partial f/\partial x = -1 \neq 0$; hence $x = h(y) = y^2$.

3) Transcendental (Lambert W). $f(x, y) = ye^y - x$. At $(0, 0)$, $\partial f/\partial y = 1$, so $y = g(x) = W(x)$ exists, analytic near 0, with $g'(x) = \Big(e^{g(x)}(1 + g(x))\Big)^{-1}$.

4) Constraint manifolds. If $\operatorname{rank} Df = m$ on $f^{-1}(0)$, the level set is an n-dimensional submanifold, locally a graph by IFT.

Failures / counterexamples.

(i) Cusp: $f(x, y) = y^3 - x^2$ at $(0, 0)$ has $\partial f/\partial y = 0$. The zero set has a cusp; it is not a graph $y = g(x)$ nor $x = h(y)$ there.

(ii) Two branches: $f(x, y) = y^2 - x$ at $(0, 0)$: cannot solve for y (two values $\pm\sqrt{x}$), but can solve for x as above. The condition must be on the variable you solve for.

(iii) Lack of smoothness: $f(x, y) = |y| - x$ at $(0, 0)$ is not C^1 in y. The zero set is $y = \pm x$, not a single branch; IFT does not apply.

Banach-space version. If E, F are Banach spaces, $f : E \times F \to F$ is C^1 near (x_0, y_0), $f(x_0, y_0) = 0$, and $D_y f(x_0, y_0) : F \to F$ is an isomorphism, then there exists a unique C^1 map g near x_0 with $f(x, g(x)) = 0$ and $Dg(x) = -\Big[D_y f(x, g(x))\Big]^{-1} D_x f(x, g(x))$.

Detailed IFT calculations: circle and parabola

Circle: $x^2 + y^2 - 1 = 0$

Let $f(x, y) = x^2 + y^2 - 1$. Then $\partial f/\partial x = 2x$, $\partial f/\partial y = 2y$.

Near $(0, 1)$ (solve $y = g(x)$). Since $\partial f/\partial y(0, 1) = 2 \neq 0$, the IFT gives a unique C^∞ function g with $g(0) = 1$ and $f(x, g(x)) \equiv 0$ for $|x|$ small (upper semicircle). The IFT derivative formula (with $n = m = 1$) yields

$$g'(x) = -\frac{\partial f/\partial x}{\partial f/\partial y}\bigg|_{(x,g(x))} = -\frac{2x}{2\,g(x)} = -\frac{x}{g(x)}.$$

Hence $g'(0) = 0$. Using the exact branch $g(x) = \sqrt{1 - x^2}$ gives $g''(0) = -1$.

Near $(1, 0)$ (solve $x = h(y)$). Here $\partial f/\partial y(1, 0) = 0$ so we cannot solve $y = g(x)$, but $\partial f/\partial x(1, 0) = 2 \neq 0$ gives a unique C^∞ function $x = h(y)$ with $h(0) = 1$ and $f(h(y), y) \equiv 0$ (right semicircle). The derivative formula (solving for x) gives

$$h'(y) = -\frac{\partial f/\partial y}{\partial f/\partial x}\bigg|_{(h(y),y)} = -\frac{2y}{2\,h(y)} = -\frac{y}{h(y)},$$

so $h'(0) = 0$. With the exact branch $h(y) = \sqrt{1 - y^2}$, one gets $h''(0) = -1$.

Parabola: $y^2 - x = 0$

Let $f(x, y) = y^2 - x$. Then $\partial f/\partial x = -1$, $\partial f/\partial y = 2y$.

At $(0, 0)$ (solve $x = h(y)$). Since $\partial f/\partial y(0, 0) = 0$, we cannot solve $y = g(x)$ there (two values $\pm\sqrt{x}$ for $x > 0$), but $\partial f/\partial x(0, 0) = -1 \neq 0$ gives a unique C^∞ function $x = h(y)$ with $h(0) = 0$. The IFT formula (solving for x) yields

$$h'(y) = -\frac{\partial f/\partial y}{\partial f/\partial x}\bigg|_{(h(y),y)} = -\frac{2y}{-1} = 2y,$$

so $h'(0) = 0$ and $h''(y) = 2$. In fact $h(y) = y^2$ exactly.

Away from $y = 0$ (solve $y = g(x)$). If $y_0 \neq 0$, then $\partial f/\partial y(x_0, y_0) = 2y_0 \neq 0$, so there is a unique branch $y = g(x)$ through (x_0, y_0) with

$$g'(x) = -\frac{\partial f/\partial x}{\partial f/\partial y}\bigg|_{(x,g(x))} = \frac{1}{2g(x)}.$$

These are the two smooth branches $y = \pm\sqrt{x}$ away from the vertex.

IFT failure details: cusp and two-branch cases

(i) Cusp: $f(x,y) = y^3 - x^2$ at $(0,0)$. We have $f_x = -2x$, $f_y = 3y^2$, so $f_x(0,0) = f_y(0,0) = 0$. The IFT hypothesis fails for both solving $y = g(x)$ and $x = h(y)$ at $(0,0)$.

The zero set is $y^3 = x^2$, i.e. $y = |x|^{2/3}$ (≥ 0), a continuous but *non-differentiable* graph over x at 0. Implicit differentiation gives $3y^2\, y' - 2x = 0$, so along the curve

$$y' = \frac{2x}{3y^2} = \frac{2x}{3|x|^{4/3}} = \frac{2}{3}\,\mathrm{sgn}(x)\,|x|^{-1/3} \xrightarrow[x\to 0]{} \pm\infty,$$

i.e. a vertical tangent and no C^1 graph $y = g(x)$ through $(0,0)$. Solving for x yields two branches $x = \pm y^{3/2}$ for $y \geq 0$; thus there is no single-valued implicit function $x = h(y)$ on a neighborhood of $(0,0)$. The zero set has a *cusp*.

(ii) Two branches: $f(x,y) = y^2 - x$ at $(0,0)$. Here $f_x = -1$, $f_y = 2y$, so $f_y(0,0) = 0$ (cannot solve $y = g(x)$: two values $y = \pm\sqrt{x}$ for $x > 0$), but $f_x(0,0) = -1 \neq 0$ so the IFT applies to solve $x = h(y)$. Indeed $x = h(y) = y^2$ with

$$h'(y) = -\frac{f_y}{f_x}\bigg|_{(h(y),y)} = -\frac{2y}{-1} = 2y, \qquad h''(y) = 2.$$

Thus the correct variable to solve for is dictated by which partial derivative is nonzero.

Example: Parabola $y^2 - x = 0$

Consider

$$f(x,y) = y^2 - x.$$

We study where the Implicit Function Theorem (IFT) allows us to express the curve as a function.

1) Evaluate the partial derivatives

$$f_x = -1, \qquad f_y = 2y.$$

2) Decide which variable can be solved for (IFT check)

- At $(0,0)$: $f_y(0,0) = 0$ (bad for $y = g(x)$), but $f_x(0,0) = -1 \neq 0$ (good for $x = h(y)$).
- At a point with $y_0 \neq 0$: $f_y(x_0, y_0) = 2y_0 \neq 0$ (good for $y = g(x)$).

3) Write the implicit function and its derivative

Case A — at $(0, 0)$: solve for x. The IFT gives a unique $x = h(y)$ with $h(0) = 0$ and $f(h(y), y) = 0$. From $y^2 - x = 0$, we get

$$\boxed{x = h(y) = y^2.}$$

Differentiate using the IFT formula (solving for the first variable):

$$h'(y) = -\frac{f_y}{f_x}\bigg|_{(h(y),y)} = -\frac{2y}{-1} = 2y, \qquad h''(y) = 2.$$

Case B — away from the vertex ($y_0 \neq 0$): solve for y. The IFT gives a local branch $y = g(x)$ through (x_0, y_0) with

$$\boxed{g'(x) = -\frac{f_x}{f_y}\bigg|_{(x,g(x))} = \frac{1}{2g(x)}.}$$

Concretely, the two branches are $y = \pm\sqrt{x}$, so on the upper one

$$g'(x) = \frac{1}{2\sqrt{x}}$$

which blows up as $x \to 0^+$.

4) If both f_x and f_y vanish

IFT does not apply (e.g. at cusps). For this parabola, that never happens on the curve except at the vertex where $f_y = 0$ but $f_x \neq 0$, so we still solve for x as $h(y) = y^2$.

Implicit representations: solving for $y = g(x)$ or $x = h(y)$

We have an implicit curve $f(x, y) = 0$.

- $y = g(x)$ means you are *solving the equation for y as a function of x*: near a point (x_0, y_0), there exists a (single-valued) map g such that

  $$f(x, g(x)) \equiv 0, \qquad g(x_0) = y_0.$$

  Geometrically, the curve is a **graph over the x-axis.**
- $x = h(y)$ means you are *solving the equation for x as a function of y*: near (x_0, y_0), there exists a (single-valued) map h with

  $$f(h(y), y) \equiv 0, \qquad h(y_0) = x_0.$$

  Geometrically, the curve is a **graph over the y-axis.**

Which one exists locally is dictated by the Implicit Function Theorem condition at (x_0, y_0):

$$\frac{\partial f}{\partial y}(x_0, y_0) \neq 0 \quad \Rightarrow \quad \text{use } y = g(x), \qquad \frac{\partial f}{\partial x}(x_0, y_0) \neq 0 \quad \Rightarrow \quad \text{use } x = h(y).$$

When they exist, their slopes are given by

$$g'(x) = -\frac{f_x}{f_y}\bigg|_{(x,g(x))}, \qquad h'(y) = -\frac{f_y}{f_x}\bigg|_{(h(y),y)}.$$

If h and g are true local inverses, then

$$h'(y_0) = \frac{1}{g'(x_0)}.$$

Tiny examples

Circle: $x^2 + y^2 - 1 = 0.$

- At $(0, 1)$: $\partial_y f = 2y = 2 \neq 0 \;\Rightarrow\; y = g(x) = +\sqrt{1 - x^2}$.
- At $(1, 0)$: $\partial_x f = 2x = 2 \neq 0 \;\Rightarrow\; x = h(y) = +\sqrt{1 - y^2}$.

Parabola: $y^2 - x = 0.$

- At $(0, 0)$: $\partial_y f = 0$ but $\partial_x f = -1 \neq 0 \;\Rightarrow\; x = h(y) = y^2$ (works),
- while $y = g(x)$ fails at the vertex (two branches $y = \pm\sqrt{x}$).

IFT via redefinition. If an equation is written as $f(x, y, x) = 0$, define

$$F(x, y) := f(x, y, x).$$

If $\det \Big(\partial_y F\Big)(x_0, y_0) \neq 0$, then by the IFT there exists a unique local function $y = g(x)$ with $F(x, g(x)) = 0$ and

$$Dg(x) = -\Big[\partial_y F(x, g(x))\Big]^{-1} \partial_x F(x, g(x)),$$

where, by the chain rule, $\partial_x F = \partial_x f(x, y, x) + \partial_3 f(x, y, x)$ and $\partial_y F = \partial_y f(x, y, x)$.

Product case. If $F(x, y) = f(x, yx)$, then

$$\partial_y F(x, y) = \partial_2 f(x, yx) \cdot x, \qquad \partial_x F(x, y) = \partial_1 f(x, yx) + \partial_2 f(x, yx) \cdot y.$$

If $\partial_y F(x_0, y_0)$ is invertible, IFT yields $y = g(x)$ near (x_0, y_0).

Implicit function for $f(x, y, z) = 0$

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be C^1 and $f(x_0, y_0, z_0) = 0$.

Solve for $y = g(x, z)$. If $f_y(x_0, y_0, z_0) \neq 0$, there exist neighborhoods and a unique C^1 map $y = g(x, z)$ with $f(x, g(x, z), z) \equiv 0$ and $g(x_0, z_0) = y_0$. The partials are

$$g_x(x, z) = -\frac{f_x}{f_y}\bigg|_{(x,g(x,z),z)}, \qquad g_z(x, z) = -\frac{f_z}{f_y}\bigg|_{(x,g(x,z),z)}.$$

Solve for $z = h(x, y)$. If $f_z(x_0, y_0, z_0) \neq 0$, there is a unique C^1 map $z = h(x, y)$ with $f(x, y, h(x, y)) \equiv 0$ and

$$h_x(x, y) = -\frac{f_x}{f_z}\bigg|_{(x,y,h(x,y))}, \qquad h_y(x, y) = -\frac{f_y}{f_z}\bigg|_{(x,y,h(x,y))}.$$

Block form (vector case). For $F : \mathbb{R}^{n+k+m} \to \mathbb{R}^m$ with variables $(u, y) \in \mathbb{R}^{n+k} \times \mathbb{R}^m$, if $F(u_0, y_0) = 0$ and $\det D_y F(u_0, y_0) \neq 0$, then there is a unique C^1 map $y = g(u)$ with $F(u, g(u)) \equiv 0$, and

$$Dg(u) = -\Big[D_y F(u, g(u))\Big]^{-1} D_u F(u, g(u)).$$

Examples

- Sphere $f = x^2 + y^2 + z^2 - 1$: at $(0, 0, 1)$, $f_z = 2 \neq 0$ so $z = g(x, y) = +\sqrt{1 - x^2 - y^2}$ with $g_x = -x/g$, $g_y = -y/g$.
- Saddle $f = x^2 - y^2 - z$: $f_z = -1 \neq 0$ so $z = g(x, y) = x^2 - y^2$; hence $g_x = 2x$, $g_y = -2y$.
- Cone $f = z^2 - x^2 - y^2$ at $(0, 0, 0)$: all partials vanish; IFT does not apply (singularity).

Solving $x = r(y, z)$ from $f(x, y, z) = 0$

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be C^1 and suppose $f(x_0, y_0, z_0) = 0$. If $f_x(x_0, y_0, z_0) \neq 0$, then by the IFT there is a unique C^1 map $x = r(y, z)$ with $r(y_0, z_0) = x_0$ and $f(r(y, z), y, z) \equiv 0$ near (y_0, z_0). Moreover,

$$r_y(y, z) = -\frac{f_y}{f_x}\bigg|_{(r(y,z),\, y,\, z)}, \qquad r_z(y, z) = -\frac{f_z}{f_x}\bigg|_{(r(y,z),\, y,\, z)}.$$

Example (sphere). For $f(x, y, z) = x^2 + y^2 + z^2 - 1$ and $(x_0, y_0, z_0) = (1, 0, 0)$ we have $f_x = 2x \neq 0$. Hence

$$x = r(y, z) = \sqrt{1 - y^2 - z^2}, \qquad r_y(y, z) = -\frac{y}{r(y, z)}, \quad r_z(y, z) = -\frac{z}{r(y, z)}.$$

Problem 17 — Iterative Tietze extension on a compact Hausdorff space

Let (X,τ) be **compact Hausdorff**, $K \subset X$ closed, and $f : K \to [-1,1]$ continuous (with the subspace topology on K).

(a) Show there is a continuous $g_1 : X \to \mathbb{R}$ with

$$\sup_X |g_1| \;\le\; \tfrac{1}{3}, \qquad \sup_K |f - g_1| \;\le\; \tfrac{2}{3}.$$

(Hint available: for closed disjoint $A, B \subset X$ and $a < b \in \mathbb{R}$, there exists a continuous "bump" $\chi : X \to [a,b]$ with $\chi \equiv a$ on A and $\chi \equiv b$ on B.)

(b) For $n \ge 1$, construct continuous $g_n : X \to \mathbb{R}$ with

$$\sup_X |g_n| \;\le\; \frac{1}{3}\left(\frac{2}{3}\right)^{n-1}, \qquad \sup_K \left| f - \sum_{j=1}^{n} g_j \right| \;\le\; \left(\frac{2}{3}\right)^{n}.$$

(c) Deduce that $g(x) = \sum_{j=1}^{\infty} g_j(x)$ defines a continuous function on X with $g|_K = f$.

Theory you need (one-liner).

- Compact Hausdorff $\Rightarrow$ normal $\Rightarrow$ **Urysohn lemma / bump functions** exist (your hint).
- A uniformly Cauchy sequence of continuous functions converges uniformly to a continuous limit on compact X.

Let (X,τ) be compact Hausdorff, $K \subset X$ closed, and $f : K \to [-1,1]$ continuous.

Solution to Problem 17

(a) Base step: coarse approximation. Let

$$A := \{x \in K : \; f(x) \ge 2/3\}, \qquad B := \{x \in K : \; f(x) \le -2/3\}.$$

Then A, B are closed and disjoint in X. By the bump-function hypothesis, there exists a continuous

$$g_1 : X \longrightarrow \left[-\tfrac{1}{3}, \tfrac{1}{3}\right] \quad \text{with} \quad g_1|_A \equiv \tfrac{1}{3}, \qquad g_1|_B \equiv -\tfrac{1}{3}.$$

Hence

$$\boxed{\sup_X |g_1| \;\le\; \tfrac{1}{3}}.$$

For $x \in K$:

$$|f(x) - g_1(x)| \le \begin{cases} 1 - 1/3 = 2/3, & x \in A, \\ 1 - 1/3 = 2/3, & x \in B, \\ |f(x)| + |g_1(x)| < 2/3, & x \in K \setminus (A \cup B), \end{cases}$$

so

$$\boxed{\sup_K |f - g_1| \;\le\; \tfrac{2}{3}}.$$

(b) Inductive refinement. Assume $g_1, \dots, g_n$ are continuous and satisfy

$$\sup_X |g_j| \le \tfrac{1}{3}\left(\tfrac{2}{3}\right)^{j-1} \qquad \text{and} \qquad \sup_K \left| f - \sum_{j=1}^{n} g_j \right| \le \left(\tfrac{2}{3}\right)^n.$$

Let the remainder on K be

$$r_n := f - \sum_{j=1}^{n} g_j, \quad \text{so} \quad |r_n| \le \left(\tfrac{2}{3}\right)^n \quad \text{on } K.$$

Apply part (a) to the scaled function $r_n/(2/3)^n : K \to [-1, 1]$ to obtain $h_{n+1} : X \to \mathbb{R}$ with

$$\sup_X |h_{n+1}| \le \tfrac{1}{3}, \qquad \sup_K \left| \frac{r_n}{(2/3)^n} - h_{n+1} \right| \le \tfrac{2}{3}.$$

Define

$$g_{n+1} := \left(\tfrac{2}{3}\right)^n h_{n+1}.$$

Then

$$\boxed{\sup_X |g_{n+1}| \;\le\; \tfrac{1}{3}\left(\tfrac{2}{3}\right)^n}, \qquad \boxed{\sup_K \left| f - \sum_{j=1}^{n+1} g_j \right| \;\le\; \left(\tfrac{2}{3}\right)^{n+1}}.$$

This completes the induction.

(c) Uniform convergence and extension. By (b) and the Weierstrass M-test,

$$\sum_{n=1}^{\infty} \sup_X |g_n| \;\le\; \sum_{n=1}^{\infty} \frac{1}{3}\left(\frac{2}{3}\right)^{n-1} \;=\; 1,$$

so $g := \sum_{n \ge 1} g_n$ converges uniformly on X, hence $g \in C(X)$. Moreover,

$$\sup_{x \in K} \left| f(x) - \sum_{j=1}^{N} g_j(x) \right| \;\le\; \left(\frac{2}{3}\right)^N \xrightarrow[N \to \infty]{} 0,$$

so $g|_K = f$. $\square$